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Information flows in trading networks[☆]Stefan J. Huber^a, Edward M. Watts^{b,c}, Christina Zhu^a*^a The Wharton School, University of Pennsylvania, United States of America^b Yale School of Management, United States of America^c European Corporate Governance Institute (ECGI), Belgium

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ABSTRACT

We study the informational value of trading networks in over-the-counter (OTC) markets. Using detailed transaction-level data from the corporate bond market, we show that investors with larger dealer networks make superior trading decisions before changes in credit fundamentals, resulting in better risk-adjusted performance. We trace these investors' superior trading decisions to trading connections where dealers are most likely to have access to novel credit-relevant information, supporting the interpretation that these investors obtain private information through their trading networks. Collectively, our evidence highlights the importance of trading relationships for investors' private information acquisition.

1. Introduction

Over-the-counter (OTC) markets play an important role in the global financial system, enabling the trading of a wide range of securities outside traditional exchanges. OTC markets are characterized by a large degree of decentralization and fragmentation. Trading requires investors to search for and bargain with dealers willing to transact, often leading to large transaction costs and low liquidity. To mitigate these frictions, investors dedicate substantial resources to building dealer networks (i.e., “trading networks”) to secure improved liquidity and pricing. However, little is known about whether these networks deliver additional benefits that would motivate investors to initiate and sustain such relationships.

Although trading networks are important for liquidity, in equilibrium they appear to be larger than would be optimal if liquidity were the only consideration. In corporate bond markets, which are among the largest and most economically important OTC markets, investors on average choose to have a *larger* dealer network despite appearing to obtain *worse* liquidity by doing

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so (Hendershott et al., 2020). Benefits offered by dealers, such as enhanced access to inventory and favorable primary-market allocations (e.g., Nikolova et al., 2020), currently are the main explanations for this empirical puzzle.

In this paper, we explore a different non-liquidity benefit—facilitating investors' information acquisition. Our paper is motivated by prior work in network economics, which highlights how economic agents can improve their outcomes by acquiring information through their networks (e.g., Jackson, 2014, 2010). More recent work on social networks in finance further highlights their importance in various financial markets settings (e.g., Hirshleifer, 2020; Kuchler and Stroebel, 2021). Hence, information is potentially an important benefit that investors can obtain from their relationships with dealers in OTC markets.

Close relationships develop in these markets because dealers compete for investors' trading flow and investors rely on dealers to transact.¹ Through these relationships, investors gain access to a wide range of private information that can complement investors' internal credit research activities.² For instance, access to formal research is typically restricted to clients via “soft dollar” arrangements where investors gain these resources in return for trading flow (e.g., Irvine, 2000; Lang et al., 2024). In addition, information from dealers' research and trading desks can informally flow to investors, through market commentary via Bloomberg chats, conversations providing “market color”, and could even raise concerns about material non-public information (MNPI) (SIFMA, 2020). As a result, larger trading networks should endow investors with more opportunities to acquire useful information. However, academic evidence on the informational benefits of trading networks is limited, largely because the opacity of OTC markets presents a substantial empirical challenge.³

We address this empirical challenge by leveraging comprehensive regulatory trading data on insurance companies, the largest institutional investor in U.S. corporate bonds (Kojien and Yogo, 2023b). These data are uniquely suited to our study, as they allow us to observe all transaction details for each of their trades, including the identities of both the insurer and the dealer. In effect, these data allow us to observe and map insurance companies' trading networks, enabling us to examine whether these networks provide informational benefits to insurers.

We begin by investigating the relation between the size of insurers' trading networks and the performance of their trades with respect to adverse credit events (i.e., downgrades and defaults). These events provide a powerful setting to detect informed trading by insurers, given they impact the value of their bond holdings (e.g., Hand et al., 1992; Cornaggia et al., 2018; Bonsall et al., 2024). Moreover, ratings directly impact their risk-based capital (RBC) charges (e.g., Kisgen and Strahan, 2010; Becker and Ivashina, 2015), which means insurers have strong incentives to hold higher-rated securities and therefore avoid these events.

Our key descriptive finding is that insurers with larger dealer networks trade more profitably ahead of adverse credit events. These results are particularly strong for those events that have the most severe consequences for bonds' valuations and insurers' risk-based capital charges. Exploiting cross-sectional variation across dealers, we also show that only connections to dealers that are more likely to have information (i.e., those with investment banking or research divisions) are associated with improved investment decisions by insurers.

We further show that this trading behavior has a real impact on insurers' portfolio performance. Specifically, comparing the risk-adjusted performance of a portfolio that emulates well-connected insurers' trading decisions to that of a similarly constructed portfolio from less-connected insurers yields monthly risk-adjusted high-low returns of approximately 7.0 basis points (t -statistic = 3.2), or approximately 0.84% annually. This outperformance is economically meaningful when considering the average monthly bond return in our sample is just 38 basis points.

While some of these performance differences can be attributed to private information sourced from their dealers, a key empirical challenge is that network size is correlated with investors' ability to access other sources of information. In particular, larger and more active insurers — who tend to have larger networks — may also be more sophisticated and have access to better resources. To better isolate the mechanism for these performance advantages, we design tests in which the information source is more likely to be outside of the insurer (i.e., with dealers). These tests directly link insurers' trading decisions to individual dealer connections, helping to rule out alternative explanations by focusing on cases in which specific dealers are the most informed about particular events. We examine two settings: dealers' security-specific information and M&A events.

First, we expect those dealers most directly associated with a bond issue to be more likely to have credit-related information useful to investors trading that bond. Specifically, dealers who originally underwrote the security are more likely to have information advantages (e.g., Chiang et al., 2019; De Franco et al., 2009; Cliff and Denis, 2004), such as through direct connections to management and covering the issuer for research purposes. Similarly, dealers who tend to be central market-makers in a bond issue are likely to be central in the flows of information on that particular security (e.g., Hagströmer and Menkveld, 2019; Babus and Kondor, 2018). We show that insurers' connections to these dealers significantly increase the propensity with which we observe selling by insurers before those bonds' downgrades, particularly junk downgrades. We also find that this propensity increases with the importance of the relationship to the dealer (i.e., whether the insurer is a top client).

¹ In practice, dealers typically assign a dedicated salesperson to manage the relationship with the institutional investor. This salesperson acts as a liaison between the investor and other parts of the dealer, facilitating investors' interactions with traders, research analysts, or the syndicate desk for primary bond issues.

² Our definition of private information is any information that may yield performance-related advantages to investors (e.g., Diamond and Verrecchia, 1987; McNichols and Trueman, 1994; Verrecchia, 2001). The information is either not publicly available or publicly available but costly to process (e.g., Blankespoor et al., 2020). This definition is broader than insider information, as it also relates to access to formal research, desk analysts, more color around trading behaviors, etc.

³ Exceptions are primarily in the equity market setting and include corporate access (e.g., Brown et al., 2015; Solomon and Soltes, 2015; Yoon, 2021), whereby brokers arrange meetings between investors and management, and broker-hosted investor conferences (e.g., Green et al., 2014; Bushee et al., 2023; Zhang, 2023).

Second, we study transactions ahead of M&A deal announcements, a powerful setting prior studies use to detect informed trading on non-public information (e.g., [Augustin et al., 2019](#); [Kacperczyk and Pagnotta, 2019](#)). We exploit the fact that private information is likely privy to the specific banks involved in the M&A transaction (e.g., [Acharya and Johnson, 2010](#); [Kang, 2024](#)). Consequently, this setting isolates the information that flows only through relationships with these banks. We find that, when an insurer is connected to a deal-affiliated dealer, the insurer is more likely to avoid losses from bonds of target companies that experience negative returns or a credit rating downgrade after the event. This relation increases with the number of potential sources of this information, as well as the importance of the relationship to the dealer. In addition to providing direct evidence of our proposed mechanism, these findings also raise questions about the leakage of MNPI, which remains an ongoing issue despite many safeguards to prevent it (see Section 2 for further discussion).

We perform several additional analyses. First, we explore the endogenous nature of network formation by examining whether a “quid pro quo” mechanism plays a role in the formation and maintenance of network connections. We expect to observe that relationships are stable and that useful information flow should lead to stronger future relationships between investors and dealers. Highlighting the stable nature of these trading relationships, we find that insurers’ *future* connections to a deal-affiliated dealer are similarly predictive of increased trading performance. Even more compelling, we find that insurers that avoid losses associated with an M&A transaction are more likely to shift future order flow to the affiliated dealers over the next twelve months, which is a form of “quid pro quo” in which insurers reward dealers with future trading flow.

Second, we examine how the information advantages of well-connected investors affect other market participants. We find evidence that the least sophisticated participants in fixed-income markets — particularly retail investors — are the most likely to incur losses.⁴ We show that, in the weeks following insurers’ sell transactions, retail investors tend to be net buyers of the same securities, highlighting that they typically take the other side of these trades. Leveraging the M&A setting again, we also show that retail investors are more likely to be net buyers in the 30 trading days before M&A announcements for target bonds that experience negative returns or credit rating downgrades after the event. These findings are in line with prior research, which documents the disadvantages of retail investors in the realm of liquidity and execution costs in fixed-income markets (e.g., [Cuny, 2018](#); [Cuny et al., 2021](#); [Griffin et al., 2023](#)).

In our final set of analyses, we rule out several additional plausible alternative explanations for our findings. First, the results are unlikely to be driven by investors splitting trades among multiple dealers to conceal private information (e.g., [Kyle, 1989](#); [Kondor and Pintér, 2022](#)). In the overwhelming majority of trading (over 95%) in a given bond on a given day, an investor transacts with only one dealer counterparty. Second, we also rule out that well-connected investors are precipitating adverse credit events through increased illiquidity ([He and Milbradt, 2014](#)). In particular, the performance differences we document are actually greatest among the most liquid securities.

Our central contribution is to provide evidence of the informational value of dealer networks. Prior studies highlight the importance of these networks for the ability of investors to obtain both liquidity and price execution (e.g., [O’Hara et al., 2018](#); [Choi et al., 2024](#); [Huber et al., 2024](#)). Significantly less is known about the non-liquidity benefits of such networks, despite suggestions that such benefits exist ([Hendershott et al., 2020](#)). We help to fill this gap and thus add to a growing body of empirical literature in accounting, economics, and finance that explores the role of networks in improving decision-making and outcomes.⁵

In studying these issues, we also contribute to the broader literature on information intermediaries in fixed-income markets. Prior research on traditional information intermediaries in these markets focus primarily on bond analysts (e.g., [De Franco et al., 2009, 2014](#); [Gillette, 2023](#)) and credit ratings agencies (e.g., [deHaan, 2017](#); [Gillette et al., 2020](#); [Hung et al., 2022](#)). Recent studies also highlight alternative intermediaries in these markets, such as fixed-income conference calls ([De Franco et al., 2023](#)), investor relations ([Kim et al., 2021](#)), and alternative data ([Kang, 2024](#)). Our paper highlights dealers as an additional information intermediary in fixed-income markets, complementing this prior literature by showing that dealers are a source through which much of the above information flows.

Finally, our paper adds to the broader literature on information acquisition and privately informed trade in fixed-income markets. While limited compared to such research in equity markets, prior research finds evidence of informed trading in many fixed-income settings, including corporate bonds (e.g., [Even-Tov, 2017](#)), credit derivatives (e.g., [Batta et al., 2016](#)), and loans (e.g., [Wittenberg-Moerman, 2008](#)). Related to our findings of informed trading before M&A activities, a subset of this literature studies the use of MNPI, with particularly strong evidence in the syndicated loan setting where syndicate participants often receive confidential borrower information (e.g., [Bushman et al., 2010](#); [Amiraslani et al., 2023](#); [Down et al., 2024](#); [Phillips, 2025](#)). Several of our findings complement this literature by suggesting that safeguards preventing the misuse of MNPI are not entirely effective as they relate to the flow of MNPI through dealer networks.

⁴ From a regulatory standpoint, retail investors are arguably the most important set of investors to consider. The SEC has repeatedly emphasized their focus on protecting retail investors in securities markets, including in fixed-income markets (e.g., [SEC, 2018, 2013](#)).

⁵ For instance, extant literature explores the various informational benefits of director networks (e.g., [Larcker et al., 2013](#); [Akbas et al., 2016](#)), analyst networks (e.g., [Martens and Sextroh, 2021](#); [Huang et al., 2022](#)), equity investor networks (e.g., [Ozsoylev and Walden, 2014](#); [Ahern, 2017](#); [Caskey et al., 2024](#); [Loh, 2026](#)), and political networks (e.g., [Christensen et al., 2017](#); [Stephan, 2021](#)). Within the context of equities markets, prior research shows evidence of leakage of informed order flow information to their best trading clients (e.g., [Barbon et al., 2019](#); [Di Maggio et al., 2019](#)).

2. Institutional details

At approximately \$11.4 trillion in principal amount outstanding, the U.S. corporate bond market is one of the world's largest and most economically important capital markets, serving as the chief source of new capital for many firms.⁶ Trading occurs in a largely decentralized marketplace characterized by fragmentation and limited price and quote transparency. To place a trade, investors need to search for a dealer willing to transact in a security. Therefore, this market is characterized by high search frictions, with dealer networks playing a key role in intermediating approximately \$59.8 billion in daily trading volume (e.g., see Bessembinder et al., 2020 for a review).

Due to the heavy involvement of dealers in these markets, investors dedicate substantial resources to developing and maintaining their dealer networks. Such relationships are essential for investors in obtaining liquidity and competitive pricing (e.g., Hendershott et al., 2020; O'Hara et al., 2018), as well as improved access to bond inventory and primary market allocations (e.g., Nikolova et al., 2020).

In addition to the above, trading relationships can also provide investors with a wide range of information about the securities they trade. For instance, access to research is typically restricted to clients via "soft dollar" arrangements where investors gain these resources in return for trading flow (e.g., Irvine, 2000; Lang et al., 2024). Highlighting these arrangements, prior studies show that both bond and equity analysts at dealers provide important information to bond market participants (e.g., Johnston et al., 2009; De Franco et al., 2009, 2014; Gillette, 2023). Outside of formal research reports, dealers' best clients can gain access to many other sources of information through dealers' research and trading desks (i.e., see Kumar et al. (2020) for related discussions for hedge fund prime brokers). For instance, through informal channels facilitated by existing trading relationships with their dealer, investors may have one-off conversations with desk analysts, receive market commentary disseminated via Bloomberg chats, or even obtain pricing information on securities.

The vast majority of this information flow is legal or occurs in regulatory "grey areas". For example, many routine practices involve arguably non-public, commercially valuable information but are still permitted, such as "market color", which SIFMA (2020) notes can "broad and fuzzy" in nature. Similarly, priority or tiered access to research is also often allowed, provided it complies with limitations on timing-based advantages and required disclosures (FINRA, 2016).

While many practices are permitted, trading on MNPI is strictly prohibited under U.S. securities law. Although the bar for convicting insider trading is high (Augustin et al., 2019), the resulting enforcement can result in severe penalties for all parties involved, including the broker-dealer itself. Therefore, broker-dealers engage in many procedures to prevent leaks of MNPI to their clients. For example, they institute information-barrier policies, including control-room oversight, restricted/watch lists, and formal wall-crossing procedures (e.g., SEC, 2012; SIFMA, 2020).

However, there is significant evidence that these guardrails are not entirely effective. For instance, prior literature finds evidence consistent with MNPI leakage before investment banking transactions in equities (e.g., Brennan et al., 2018), options (e.g., Lowry et al., 2019), and credit derivatives (Acharya and Johnson, 2007). Evidence of ethical wall breakdowns has previously been documented among equity analysts (Chen and Martin, 2011), private loan contracting (Ivashina and Sun, 2011), prime brokerages (Kumar et al., 2020), and in the context of order flow leakage (Barbon et al., 2019).

In practice, fully eliminating leaks of MNPI is particularly difficult in our institutional setting for several reasons. Regulators note that resource constraints prevent them from investigating every case (FINRA, 2008). They rationally prioritize enforcement on markets or instruments in which abnormal profits are larger or those which generate the greatest amount of harm (e.g., SEC, 2017). Fixed-income securities are generally less price-sensitive and dominated by sophisticated institutional investors rather than retail investors, whom the SEC explicitly prioritizes protecting. These features explain why, anecdotally, fixed-income enforcement teams tend to be smaller. As a result, enforcement activity appears lower in corporate fixed-income markets, although exceptions exist.⁷ Given prior evidence highlighting strong associations between enforcement activities or resources and trading on MNPI (e.g., Kim et al., 2025; Kacperczyk and Pagnotta, 2024; Nguyen, 2021), leakage of MNPI is a distinct possibility in our setting.

Additionally, the nature of the information flows within corporate bond markets can make compliance challenging. Given the frequency of primary offerings, pre-transaction investor outreach ("sounding the market") is common, and syndicate desks work closely with both sales and trading units (e.g., SEC, 2012; SIFMA, 2020). Broker-dealers are required to implement controls and documentation for the exchange of these types of information. Secondary corporate bond trading is also less naturally organized around contractually specified MNPI access rights, such as those found in syndicated loan markets (Amiraslani et al., 2023), and more around relationship-based communication. These communications can be informal and documentation/audit trails may be incomplete, which makes monitoring for misuse of MNPI difficult (SEC, 2012). These institutional details highlight the potential for MNPI leaks, which is further emphasized by continued enforcement activities in these areas of broker-dealer and investment advisor failures around MNPI (e.g., SEC, 2024b).

⁶ All numbers in this section are calculated by the Securities Industry and Financial Markets Association (SIFMA) for Q2 of 2025. See: <https://tinyurl.com/2u2kmwnd>.

⁷ For example, the SEC charged a bond salesman and hedge fund manager for insider trading in 2009 (SEC, 2009). More recently, it charged an investment advisor for failing to enforce the proper guardrails to prevent the misuse of MNPI in distressed debt markets (SEC, 2024a).

3. Data

3.1. Transaction and dealer trading network data

We develop our dealer trading network data based on comprehensive transaction-level data on insurance companies' trading in corporate bonds. Although trading networks are typically unobservable in other settings, an advantage of our setting is that U.S.-based insurance companies are required to file a quarterly and annual Schedule D with the National Association of Insurance Commissioners (NAIC). This schedule includes detailed information on each transaction, including the transaction date, the direction of the trade, the quantity traded, and the price paid. Vital to our analyses, it provides the identities of the insurance company and the dealer with whom each trade is completed.⁸

We obtain annual Schedule D data for all acquisitions and disposals of corporate bonds by all U.S. insurance companies (e.g., health, life/fraternal, and property and casualty insurance companies) from 2009 to 2022 from S&P Capital IQ's Insurance Investment Transactions database.⁹ S&P Global aggregates these Schedule D filings into a structured format for investors and other S&P clients, and performs cleaning and data quality assessment. We follow O'Hara et al. (2018) and implement several further cleaning and filtering steps. Specifically, we retain transactions only for bonds whose face value is standard (i.e., \$1000), issuance amounts of \$1 million or greater, and those with characteristics data on Mergent FISD (e.g., only U.S. bonds). We also follow the steps that the paper outlines to correct potential data errors related to reported prices.

We implement several additional cleaning techniques. To focus on secondary market transactions, we drop those trades occurring in the 60 days after issuance and those related to non-secondary market transactions, such as bond redemptions, maturities, and primary market issuances (e.g., Hendershott et al., 2020). Importantly, we also clean and standardize the dealer names. The filings contain self-reported counterparty names, which can include typos and different variations of reported names for the same dealer. We clean dealer names using a combination of manual cleaning, standardizing common abbreviations and suffixes, and fuzzy matching. We further consolidate all insurer trading data at the Ultimate Parent level.¹⁰ The resulting dataset consists of approximately 2.0 million transactions across 1577 insurers and 370 dealers.

We then construct the dealer network of each insurer, which enables us to measure both the number of separate counterparties and the total volume transacted between counterparties in our dataset. Fig. 1 shows three example insurers' dealer networks in 2018. It depicts the dealer connections for an insurance company with few connections (Hoosier Motor Club), an average number of connections (Grange), and a high number of connections (Independence). The width of each connection edge in the network corresponds to the fraction of the insurer's trading volume that was executed with the respective dealer in 2018.

Fig. 2 visually depicts examples of dealer trading networks using pie charts to represent the distribution of each brokerage firm's trading volume across trading clients who are insurers. The figure focuses on two example brokerage firms' trading in 2019. Panel A shows the insurer trading client base of Goldman Sachs. For illustrative purposes, the figure breaks out the top 20 insurer trading clients. Goldman's top 20 insurers by trading volume make up more than half of its total corporate bond trading volume with insurers. Panel B shows the insurer trading client base of Nomura. Nomura's top 20 insurers by trading volume make up more than 75% of its total corporate bond trading volume with insurers. In both panels, there is significant variation in the strength of the trading connection even among the dealer's top 20 insurer trading clients.

Table 1 presents summary statistics of the dealer networks in our sample. Panel A presents measures of insurance companies' dealer networks. For each insurer-year, we obtain the insurer's trading volume, number of trades, and number of dealers with whom they transact. The average trading volume across all insurer-years is \$266.027 million. On average, insurers execute 163.886 trades across 12.785 dealers per year.

Panel B of Table 1 presents measures of dealers' insurer trading networks. For each dealer-year, we obtain the dealer's trading volume, number of trades with insurers, and the number of insurers with whom they transact. The average trading volume with insurers across all dealer-years is \$1.314 billion. On average, dealers execute 774.265 trades across 58.650 insurers per year.

We also report measures of the strength of the trading connection between insurers and dealers. Panel C of Table 1 provides descriptive statistics of the connection strength at the dealer-insurer-year level. Trading volume with each insurer, on average, represents 1.359% of the total trading volume of a brokerage during the year (% of Dealer Volume). Trading volume with each brokerage, on average, represents 7.796% of the total trading volume of an insurer during a year (% of Insurer Volume).

3.2. Transaction-level sample

Our sample selection process for the transaction-level analyses begins with the sample of approximately 2.0 million bond transactions described in Section 3.1. To study differences in decision-making based on trading connections, we compute each insurance company's number of unique dealer connections in the rolling 12-month window before the transaction. We require future credit ratings data from Mergent FISD to compute whether the bond is downgraded in the future (i.e., transactions for bonds that are not rated in the month of interest after the transaction are dropped). We also require data to compute control variables: the rating

⁸ These records are comprehensive and highly accurate. The signature of an insurance company officer (e.g., president or comptroller) is required, and inaccuracies or fraudulent reports can lead to statutory fines or criminal charges.

⁹ The sample period begins in 2009, when S&P Global began cleaning and aggregating these data from Schedule D filings (e.g., Hagenberg, 2025).

¹⁰ For example, if we see trades by insurance companies A and B, both of which are owned by parent company P, we assign all trades to P. For ease of exposition, we use the terms "insurer", "insurance company", and "investor" interchangeably to refer to the parent company.

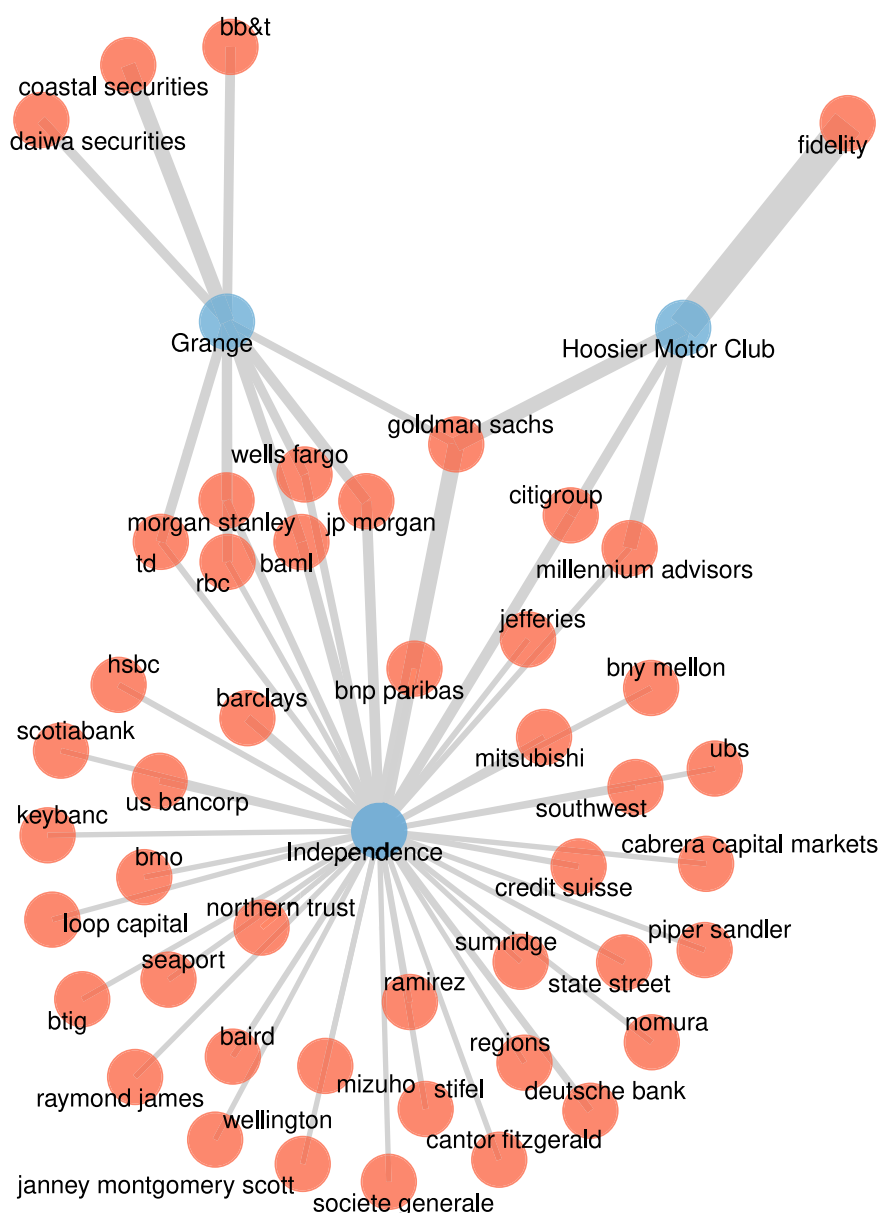


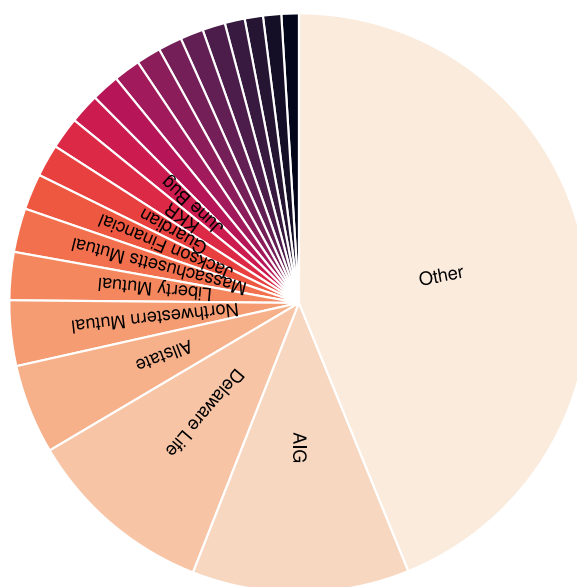
Fig. 1. Examples of insurers' dealer networks.

This figure presents the dealer networks of three insurers (blue), Grange, Independence, and Hoosier Motor Club, with their dealers (orange) in 2018. The edge widths in the network correspond to the fraction of the insurer's trading volume executed with the respective dealer in 2018. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

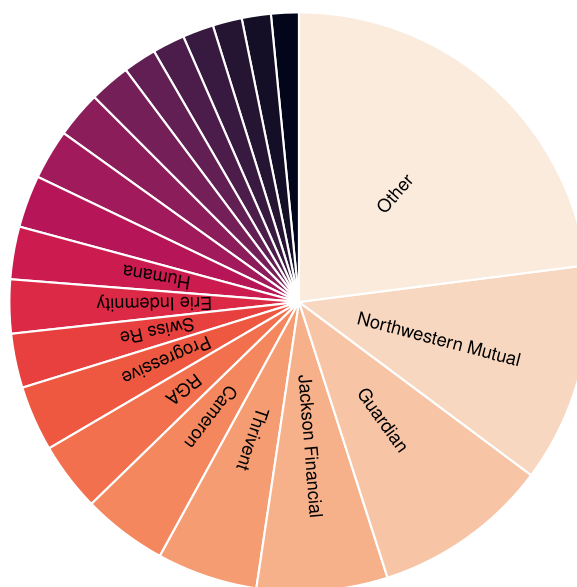
level of the bond, the remaining time-to-maturity, and the natural logarithm of the bond offering amount (all from Mergent FISD). After these data requirements, our final transaction-level sample consists of 1,598,694 transactions.¹¹ The first panel in [Appendix A](#) defines all variables used in the transaction-level analyses.

Panel A of [Table 2](#) presents summary statistics for the main variables we use in our transaction-level analyses. The average *Sell* is 0.45, indicating that the secondary market buy-to-sell ratio is approximately $1.22 (\approx 0.45 / (1 - 0.45))$. On average, insurance companies have more buy than sell transactions, highlighting the buy-and-hold nature of their investment strategies. We also see

¹¹ This sample measures downgrades in the next 3 months. The sample size is slightly reduced when we use specifications that require measures computed over a longer horizon after the trade (e.g., downgrades over a 12-month window in the future). The sample size slightly increases when we use specifications that examine defaults, which do not require data on future downgrades.



(a) Goldman Sachs.



(b) Nomura.

Fig. 2. Dealer connections examples.

This figure presents the distribution of intermediated trading volume for two example brokerage firms across each brokerage's trading clients who are insurers in 2019. Panel (a) shows the insurer customer base of Goldman Sachs, and Panel (b) shows the insurer customer base of Nomura. For illustrative purposes, the top 20 connections, ranked based on the insurer's % of the brokerage's trading volume, are separately broken out. The remaining connections are aggregated in the *Other* category.

the average transaction is conducted by an insurer with \$11.1 billion in corporate bond holdings, and which turns over about 20% of its AUM in trading each year. Panel A also reports the other variables we use in our transaction-level analyses, all of which are in line with expectations and prior studies.

3.3. Event-level sample

In addition to our transaction-level analyses, we perform several analyses at the event level around two sets of events, downgrades and M&A announcements. We consider the universe of all downgrade events over our sample period. To avoid double-counting

Table 1
Network summary statistics.

Panel A: Insurer networks								
	Mean	StDev	p ^{5%}	p ^{25%}	p ^{50%}	p ^{75%}	p ^{95%}	Obs.
Trading Volume (\$MM)	266.03	792.67	0.24	2.13	12.68	89.07	1712.81	14,851
No. of Trades	163.89	345.23	2.00	12.00	37.00	129.00	825.00	14,851
No. of Dealers	12.79	11.98	1.00	2.00	10.00	20.00	37.00	14,851
Panel B: Dealer networks								
	Mean	StDev	p ^{5%}	p ^{25%}	p ^{50%}	p ^{75%}	p ^{95%}	Obs.
Trading Volume (\$MM)	1314.309	4858.56	0.50	6.02	43.14	305.405	5802.424	3229
No. of Trades	774.27	2631.79	1.00	14.00	54.00	282.00	3717.40	3229
No. of Insurers	58.65	129.41	1.00	1.00	3.00	30.00	376.60	3229
Panel C: Dealer-insurer connection strength								
	Mean	StDev	p ^{5%}	p ^{25%}	p ^{50%}	p ^{75%}	p ^{95%}	Obs.
% of Dealer Volume	1.36	6.48	0.00	0.01	0.06	0.38	3.77	190,560
% of Insurer Volume	7.80	15.92	0.07	0.76	2.87	8.07	26.96	190,560

This table reports summary statistics of the annual dealer trading networks of insurance companies. Panel A and Panel B present descriptives at the insurer-year level and the dealer-year level, respectively. Panel C provides descriptive statistics of the connection strength at the dealer-insurer-year level. Panel C presents both the fraction of the total trading volume of a dealer that is with the insurer during the year (% of Dealer Volume) or the fraction of the total trading volume of an insurance company that is with a dealer in a year (% of Insurer Volume), both measured in percentage points. All variables are winsorized at the 1st and 99th percentiles.

events, we retain the first downgrade across S&P, Moody's, and Fitch, following [Ellul et al. \(2011\)](#). We collect M&A event data from Refinitiv (formerly SDC Platinum). There are 1327 deals larger than \$5 million where the target company has bonds outstanding at the time of the deal announcement (e.g., [Beneish et al., 2022](#)).

We apply a series of filters to focus our sample on deals likely to be informationally relevant for the target company's bonds. We require the acquirer to obtain a controlling stake (i.e., more than 50% of shares in the target). We further exclude divestitures, spin-offs, split-offs, and acquisitions as part of a bankruptcy process. This sample selection procedure leaves us with a final sample of 540 M&A deals.

For both downgrade and M&A events, we create a dataset of trading opportunities at the insurer-event level. Specifically, we examine the trading activity of all insurers in the downgraded bond or the bonds of the acquired company in the 30 trading days before the event. This process results in a sample of 87,655,544 and 854,789 insurer-event observations for all downgrades and M&A events, respectively.

We then determine whether an insurer has an informationally relevant trading connection for a specific event. In the downgrade sample, we consider connections to the underwriter and specialist market-makers of a downgraded bond. We retrieve underwriter information from Mergent FISD and identify specialist market-makers from our transaction data, defining the specialist as the dealer intermediating the largest trading volume in a particular bond. In the M&A sample, we identify the investment banks that advised either the target or the acquirer of a deal and link them to their respective dealer arms in our trading network data. Using these data, we measure an insurance company's trading connections with an underwriter, specialist, or deal advisor over the 12 months prior to the event.

Panels B and C of [Table 2](#) cover variables measured at the insurer-event level, which we use for our analyses of the relation between insurers' connections to dealers and their trading activity around specific credit events. The third panel in [Appendix A](#) defines all variables used in the insurer-event level analyses. For each insurer-event, we compute their aggregate buy-sell imbalance in the 30 trading days before the event (*Insurer BSI*) as the difference between the insurer's buy volume and the insurer's sell volume, scaled by the sum of the insurer's buy and sell volume and multiplied by 100, and set to zero if the insurer has no trading volume in those bonds. In Panel B, the average *Insurer BSI* is -0.097% (i.e., approximately neutral). On average, 13.3% (25.6%) of insurers have a trading relationship with a specialist (underwriter) dealer during the 12-month window prior to the downgrade. On average, 0.027% (0.040%) of the specialist (underwriter) dealer's trading volume is with each insurer (*Specialist Connection Strength* and *Underwriter Connection Strength*, respectively). Finally, 4.1% of the downgrade events are downgrades to junk status.

In Panel C, the average *Insurer BSI* is -0.027% (i.e., indicating more sell volume but still close to neutral), which is driven by the sparse trading of insurance companies around these events. In untabulated analyses, we condition on insurers that trade in the 30 days before the event (i.e., dropping the *Insurer BSI* that is 0 due to no trading volume) and find that sells outweigh buys ahead of M&A events in a ratio of 5-to-3. Given that these events are generally perceived as negative based on the subsequent negative bond returns, these statistics imply that insurers that do trade before these events appear to be relatively informed, on average.

On average, 36.7% of insurers have a trading relationship with a deal-affiliated dealer during the 12-month window prior to the deal announcement. The average number of unique deal-affiliated dealers with which insurers have trading relationships is 0.695, and on average, 0.066% of a deal-affiliated dealer's trading volume is with each insurer (*Deal Connection Strength*). The average M&A event return for target company bonds is -4.683% , with most (i.e., 76.5%) of the events accompanied by negative returns. Finally, 2.2% of the events are followed by downgrades for target company bonds within the next 3 months.

Table 2
Sample summary statistics.

Panel A: Transaction level summary statistics								
	Mean	StDev	p ^{5%}	p ^{25%}	p ^{50%}	p ^{75%}	p ^{95%}	Obs.
Sell	0.45	0.50	0.00	0.00	0.00	1.00	1.00	1,560,073
No. of Dealers	31.75	14.70	2.00	23.00	33.00	42.00	54.00	1,560,073
Downgrade ^{3mo.}	2.86	16.66	0.00	0.00	0.00	0.00	0.00	1,540,829
Junk Downgrade ^{3mo.}	0.23	4.81	0.00	0.00	0.00	0.00	0.00	1,540,829
Default ^{1Y}	0.19	4.41	0.00	0.00	0.00	0.00	0.00	1,560,073
Time to Maturity (Years)	8.50	7.37	1.21	3.97	6.66	9.84	28.556	1,560,073
Bond Size (\$BN)	1.04	0.82	0.25	0.50	0.75	1.25	2.99	1,560,073
Rating	8.45	3.11	4.00	6.00	8.00	10.00	14.00	1,560,073
AUM (\$BN)	11.13	20.05	0.02	0.38	2.29	11.38	63.35	1,560,073
Yearly Trade Volume (\$BN)	2.17	3.47	0.01	0.14	0.70	2.96	9.749	1,560,073
Panel B: Insurer-Downgrade event level summary statistics								
	Mean	StDev	p ^{5%}	p ^{25%}	p ^{50%}	p ^{75%}	p ^{95%}	Obs.
Insurer BSI	-0.10	6.41	0.00	0.00	0.00	0.00	0.00	87,655,544
Specialist Connection	0.13	0.34	0.00	0.00	0.00	0.00	1.00	87,655,544
Specialist Connection Strength	0.03	0.40	0.00	0.00	0.00	0.00	0.038	87,655,544
Underwriter Connection	0.26	0.44	0.00	0.00	0.00	1.00	1.00	87,655,544
Underwriter Connection Strength to Junk	0.04	0.18	0.00	0.00	0.00	0.00	0.171	87,655,544
	0.04	0.20	0.00	0.00	0.00	0.00	0.00	87,655,544
Panel C: Insurer-M&A event level summary statistics								
	Mean	StDev	p ^{5%}	p ^{25%}	p ^{50%}	p ^{75%}	p ^{95%}	Obs.
Insurer BSI	-0.03	4.42	0.00	0.00	0.00	0.00	0.00	854,789
Deal Connection	0.37	0.48	0.00	0.00	0.00	1.00	1.00	854,789
# Deal Connections	0.70	1.12	0.00	0.00	0.00	1.00	3.00	854,789
Deal Connection Strength	0.07	0.26	0.00	0.00	0.00	0.01	0.34	854,789
Event Return	-4.68	8.50	-23.61	-6.48	-1.64	-0.07	2.33	486,044
Negative Return	0.77	0.42	0.00	1.00	1.00	1.00	1.00	486,044
Downgrade ^{3mo.}	0.02	0.15	0.00	0.00	0.00	0.00	0.00	854,789

This table reports descriptive statistics of observations used in the study. Panel A presents descriptive statistics of variables used in our transaction-level analyses. For ease of interpretation, we report the raw, unranked version of the variable *Connection Rank*, which is labeled “No. of Dealers”. The logged variables Bond Size, AUM, and Yearly Trading Volume are presented unlogged. Panel B presents descriptive statistics of variables measured at the insurer-event level for the downgrade events. Panel C presents descriptive statistics of variables measured at the insurer-event level for the M&A events. In addition to descriptive statistics for the variable *Negative Return*, we also report the continuous event return, which is labeled “Event Return”.

4. Dealer network size and trading performance

This section explores the association between insurers’ dealer network size and their trading performance. We begin by examining the relation between insurers’ dealer network size and trading activity in anticipation of upcoming adverse credit events. We then use portfolio analyses to assess the aggregate impact of the improved decision-making of those insurers with more extensive dealer networks.

4.1. Dealer network size and future adverse credit events

We begin by exploring the relation between the size of insurers’ dealer networks and the informativeness of their trading decisions for future credit rating downgrades. Credit rating downgrades provide a powerful setting for exploring the impact of network size on decision-making with respect to trading performance for two reasons. First, they significantly impact bond values and, therefore, directly impact insurer companies’ portfolios (e.g., [Hand et al., 1992](#); [Cornaggia et al., 2018](#); [Bonsall et al., 2024](#)). Second, insurers have strong incentives to hold higher-rated securities, given the ratings directly impact their risk-based capital (RBC) charges (e.g., [Kisgen and Strahan, 2010](#); [Becker and Ivashina, 2015](#)).¹²

To explore the above, we estimate variations of the following linear probability model using our sample of insurer transactions described in Section 3.2:

$$\begin{aligned} \text{Downgrade}_{b,t+h} = & \beta_1 \text{Sell}_{i,b,t} \times \text{Connection Rank}_{i,t} + \beta_2 \text{Connection Rank}_{i,t} + \beta_3 \text{Sell}_{i,b,t} \\ & + \gamma \text{Controls}_{i,b,t} + \sum \beta_r \text{Rating}_{r,b,t} + \sum \beta_t \text{Year} \times \text{Month}_t + \sum \beta_i \text{Insurer}_i \end{aligned}$$

¹² The RBC requirement is a statutory minimum level of capital based on the riskiness of its financial assets and its underwriting risk ([Kojien and Yogo, 2023a](#)). The formula to calculate the exact amount of required capital is complex, but as the RBC bond factor increases, so will the required risk-based capital (i.e., see [Appendix B](#)).

$$+ \sum \beta_b \text{Bond}_b + \epsilon_{i,b,t+h}, \quad (1)$$

where $\text{Downgrade}_{b,t+h}$ is one of several indicators for whether the bond b has a rating downgrade in the next $h = 3$ or 12 months after transaction date t . Sell is an indicator variable set to one if the transaction is a sell transaction and zero if it is a buy transaction. Connection Rank is the quintile rank of the number of dealers with whom the insurer traded in the 12-month window prior to the transaction date.¹³

Our coefficient of interest, β_1 , measures how insurers' sell transactions' ability to predict future ratings downgrades varies with their network size, relative to their buy transactions'. A positive β_1 indicates that insurers with more extensive dealer networks are more likely to sell, rather than purchase, securities before upcoming credit rating downgrades. Said differently, it implies insurers' network size influences their ability to improve their trading performance by better predicting future credit rating downgrades.

To control for potential differences in insurers' bond portfolios and resources at their disposal, we include a vector of controls (*Controls*) for bond characteristics, including the remaining time-to-maturity and the bond size (natural logarithm of the bond offering amount), and insurer characteristics, including the insurer's bond portfolio size (natural logarithm of par value) and the insurer's trading activity (natural logarithm of par value traded).¹⁴ We also include fixed effects indicating each round credit rating level r . Our primary and most stringent specification also includes year-month t fixed effects, insurer i fixed effects, and bond b fixed effects. Insurer fixed effects account for time-invariant insurance company characteristics, such as an insurer's time-invariant in-house research capabilities. More than 25% of the insurers in our sample change *Connection Rank* during the sample period. The inclusion of bond fixed effects holds constant securities in which insurers trade, only capturing time dynamics in insurers' abilities to predict downgrades.

Panel A of Table 3 presents regression estimates for multiple variations of Eq. (1).¹⁵ Our results show that the transactions of well-connected insurers are more informative about future credit rating downgrades. Specifically, across nearly all specifications, we find positive and statistically significant coefficients on β_1 , implying that the sell transactions of well-connected insurers (i.e., those with more trading connections in the past year) are relatively more positively associated with the probability that the bond has a future downgrade than the sell transactions of less-connected insurers. These estimates are statistically significant at conventional levels, with the exception of column 2 (the analyses with the least amount of statistical power).

In addition to being statistically significant, our findings are also economically significant. For instance, column 4 suggests that the sales of insurers with the largest dealer network (quintile 5) are approximately 31% more likely to be followed by a downgrade over the next twelve months than those with the smallest dealer network (quintile 1).¹⁶ Moreover, column 4 includes bond fixed effects, highlighting that well-connected insurers sell at the most important times within the bond's life. This finding rules out many alternative explanations, such as well-connected insurers' ability to purchase better-performing bonds.

We provide further evidence of the relation between insurers' dealer network size and trading decisions by leveraging the fact that more severe downgrades are significantly more important for insurers to avoid in their holdings. In particular, insurers have strong regulatory incentives to hold investment-grade securities (e.g., Becker and Ivashina, 2015), which is highlighted by the fact that $\approx 95\%$ of total bonds held by insurers at the end of 2022 were investment-grade bonds (NAIC, 2022). Avoiding downgrades to non-investment-grade (i.e., "junk") is also particularly important, in part because they may lead to forced fire sales at a loss (Ellul et al., 2011). At the extreme, defaults result in substantial dollar losses.¹⁷

Motivated by the above, we investigate the relation between the size of insurers' dealer networks and the propensity to avoid future downgrades to non-investment-grade status and future defaults. To do so, we re-estimate multiple variations of Eq. (1) that replace the dependent variable with an indicator that the bond is downgraded to junk status or has a future default over various measurement windows. Panel B of Table 3 presents our estimates for junk downgrades (columns 1 and 2) and defaults (columns 3 and 4).

Across all columns, our findings show that well-connected insurers' bond sale transactions, relative to their bond purchases, better predict both junk downgrades and defaults. Specifically, the coefficients on $\text{Sell} \times \text{Connection Rank}$ are positive and statistically significant at the 1% level or lower threshold. Moreover, the economic magnitudes increase with event severity. In terms of standard deviations, the effect is twice (1.6 times) as large for junk downgrades (defaults) as for regular downgrades. These findings confirm that insurers with larger dealer networks appear to avoid adverse credit events in cases where they have the strongest incentives to do so.

¹³ We use the quintile rank for ease of interpretation. Untabulated results using the number of dealers are qualitatively similar.

¹⁴ Many of the characteristics associated with insurers' sophistication and information acquisition resources are unobservable (e.g., their diligence teams). Our regressions control for observable measures, insurer size and trading activity, which proxy for the various resources at the insurer's disposal, such as the size of their diligence team. Untabulated analyses show that these measures are positively correlated with and can explain up to 55% of the variation in insurers' dealer network size.

¹⁵ An inherent advantage of the longer measurement window is statistical power, because there is substantially more variation in the propensity to observe credit rating downgrades over longer periods. Another advantage is that it reduces the possibility we are capturing insurers' trading in response to credit rating changes from another credit rating agency. Specifically, although we focus on S&P rating changes, insurers in our sample may subscribe to other rating agencies, such as Egan Jones, which might be timelier in some cases (Beaver et al., 2006, see Bonsall et al., 2017 for arguments otherwise).

¹⁶ Note that the baseline is 0.514 for the 12-month horizon for the lowest quintile. Moving to the highest quintile adds 0.16 (i.e., $4 \times (0.12 - 0.08) \approx 0.16$).

¹⁷ For instance, Moody's Investor Service (2024) calculates that the trailing five-year recovery values for senior unsecured debt, which comprises the vast majority of insurers' corporate bond holdings, were just 35.7 cents on the dollar (i.e., an approximate 64.3% loss from initial issuance values of par).

Table 3
Investors' connections and adverse credit events.

Panel A: Downgrades				
	Downgrade ^{3mo.}		Downgrade ^{12mo.}	
	(1)	(2)	(3)	(4)
Sell × Connection Rank	0.136*** (2.973)	0.048 (1.238)	0.280*** (2.853)	0.120* (1.842)
Connection Rank	−0.266*** (−6.134)	0.000 (−0.009)	−0.715*** (−7.184)	−0.080 (−1.324)
Sell	0.474** (2.401)	0.523*** (2.984)	0.871** (2.055)	0.514* (1.756)
Controls	Yes	Yes	Yes	Yes
Rating fixed effects	Yes	Yes	Yes	Yes
Year × Month fixed effects	No	Yes	No	Yes
Insurer fixed effects	No	Yes	No	Yes
Bond fixed effects	No	Yes	No	Yes
Observations	1,540,829	1,540,829	1,470,043	1,470,043
R ²	0.008	0.208	0.021	0.430
Panel B: Severe credit events				
	Junk Downgrade ^{3mo.}		Default ^{1Y}	
	(1)	(2)	(3)	(4)
Sell × Connection Rank	0.037*** (4.180)	0.040*** (4.558)	0.026*** (3.688)	0.024*** (4.724)
Connection Rank	−0.043*** (−6.341)	−0.023*** (−3.117)	−0.017*** (−2.982)	−0.004 (−0.808)
Sell	0.028 (0.817)	−0.054 (−1.471)	−0.001 (−0.044)	−0.072*** (−3.393)
Controls	Yes	Yes	Yes	Yes
Rating fixed effects	Yes	Yes	Yes	Yes
Year × Month fixed effects	No	Yes	No	Yes
Insurer fixed effects	No	Yes	No	Yes
Bond fixed effects	No	Yes	No	Yes
Observations	1,540,829	1,540,829	1,560,073	1,560,073
R ²	0.032	0.172	0.132	0.662

This table presents analyses that examine the effect of insurers' network connections on the predictive ability of their bond sales for future adverse credit events. We estimate several versions of Eq. (1) using the sample of transactions described in Section 3. Panel A presents results for downgrades that happen within 3 or 12 months of the trade. Panel B presents results for downgrades to junk status that happen within 3 months of the trade or defaults that happen within 1 year of the trade. *Controls* include Time to Maturity, Bond Size, AUM, and Yearly Trade Volume. All variables are defined in Appendix A. *t*-statistics based on standard errors clustered by insurer are in parentheses. Levels of significance are presented as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

4.2. Cross-sectional analyses

Our findings thus far highlight that insurers who trade with more dealers appear to be more informed. To provide initial evidence that investors with large dealer networks are obtaining information advantages from dealers themselves, rather than from unobservable investor characteristics correlated with network size (e.g., a larger diligence team), we explore cross-sectional variation across dealers. Specifically, we exploit the heterogeneity across dealers in their likelihood of having novel information that would be useful to investors, as measured by the additional functions certain dealers have.

To explore the above, we re-estimate Equation (1), replacing the network size variable with two separate variables measuring the number of investors' trading connections to informed and uninformed dealers. These dealers are those expected to be more or less likely to have value-relevant information, respectively. We identify informed dealers in two ways. First, we identify dealers with an investment banking arm, as measured by whether they advised at least one M&A deal in the Thomson Reuters SDC database. Second, we identify dealers with research arms, as measured by whether they have at least one research report filed on IBES.¹⁸

Panel A of Table 4 reports our estimates for these regressions when using splits based on whether a dealer has been active in the M&A market. Across all columns, we find a positive and statistically significant coefficient on the interaction *Sell* × *M&A Connection Rank*, consistent with a positive effect of investors' connections with M&A-affiliated dealers on the predictive

¹⁸ We focus on equity research arms rather than fixed-income for several reasons. Data limitations make identifying the full universe of trading desks challenging, and single-name coverage for individual bond securities is rare, particularly for the high-grade bonds that insurers hold (e.g., De Franco et al., 2009). Additionally, discussions with industry practitioners confirm that equity research analysts are an important source of information for fixed-income investors in credit markets.

Table 4
Cross-sectional analyses: Heterogeneity in dealer network functions.

Panel A: Dealer M&A transaction activity				
	Downgrade ^{3mo.} (1)	Downgrade ^{12mo.} (2)	Junk Downgrade ^{3mo.} (3)	Default ^{1Y} (4)
Sell × M&A Connection Rank	0.120** (2.110)	0.307*** (3.471)	0.034*** (2.811)	0.035*** (4.463)
Sell × Non-M&A Connection Rank	−0.086* (−1.672)	−0.194** (−2.319)	0.008 (0.646)	−0.011 (−1.279)
Controls	Yes	Yes	Yes	Yes
Rating fixed effects	Yes	Yes	Yes	Yes
Year × Month fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	Yes
Observations	1,540,829	1,470,043	1,540,829	1,560,073
R ²	0.208	0.430	0.172	0.662
Panel B: Dealer research activity				
	Downgrade ^{3mo.} (1)	Downgrade ^{12mo.} (2)	Junk Downgrade ^{3mo.} (3)	Default ^{1Y} (4)
Sell × Research Connection Rank	0.056 (0.829)	0.193* (1.905)	0.038** (2.402)	0.039*** (4.227)
Sell × Non-Research Connection Rank	−0.024 (−0.372)	−0.069 (−0.691)	0.003 (0.181)	−0.013 (−1.328)
Controls	Yes	Yes	Yes	Yes
Rating fixed effects	Yes	Yes	Yes	Yes
Year × Month fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	Yes
Observations	1,540,829	1,470,043	1,540,829	1,560,073
R ²	0.208	0.430	0.172	0.662

This table presents analyses that examine the effect of insurers' network connections to different types of dealers on the predictive ability of their bond sales for future adverse credit events. We estimate several versions of Eq. (1) using the sample of transactions described in Section 3. Panel A ranks insurers separately based on their network connections with M&A deal-advising and non-M&A deal advising dealers. Panel B ranks insurers separately based on their network connections with dealers that have and that do not have a research practice. All regressions include main effects, which are omitted from the output for parsimony. Controls include Time to Maturity, Bond Size, AUM, and Yearly Trade Volume. All variables are defined in Appendix A. *t*-statistics based on standard errors clustered by insurer are in parentheses. Levels of significance are presented as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

ability of insurers' sell transactions (relative to their buy transactions) for future downgrades, downgrades to junk, and defaults. In contrast, the coefficient on the interaction *Sell × Non-M&A Connection Rank* is insignificant in all columns except columns 1 and 2, where it is negative and significant or marginally significant.

Panel B of Table 4 reports the results of these regressions when using splits based on whether a firm has a research division. We find a positive and significant coefficient on the interaction *Sell × Research Connection Rank*, consistent with a positive effect of insurers' connections with research-affiliated dealers on the predictive ability of investors' sell transactions (relative to their buy transactions) for future downgrades (columns 1 and 2), downgrades to junk (column 3), and defaults (column 4). In column 1, when we investigate future 3-month downgrades, the coefficient on the interaction term is insignificant (but we note that the coefficient is positive and marginally significant using 12-month downgrades in column 2). In contrast, the coefficient on the interaction *Sell × Non-Research Connection Rank* is insignificant in all columns. Overall, our findings indicate that the predictive abilities of well-connected insurers prior to adverse credit events are driven by their connections to more informed dealers.

4.3. Aggregate portfolio analyses

The findings in the prior subsections highlight that insurers with larger dealer networks make more informed trading decisions regarding future adverse credit events. In this section, we investigate the aggregate economic effect of the improved decision-making by well-connected insurers on their portfolio performance. To provide evidence on this issue, based on similar analyses in Di Maggio et al. (2019), we calculate the risk-adjusted abnormal performance of insurers with large dealer networks and compare it to that of insurers with smaller dealer networks.

We obtain monthly bond returns data from the Wharton Research Data Services (WRDS) Bond Returns database.¹⁹ Using these data, at the beginning of each month, we construct long-short value-weighted portfolios that purchase (sell) bonds in the highest

¹⁹ These data are increasingly used in academic research to explore a variety of corporate bond-related research questions (e.g., Li, 2022; deHaan et al., 2023; Gad et al., 2024). A detailed description of the data, cleaning steps, and associated code can be found on WRDS: <https://wrds-www.wharton.upenn.edu/pages/get-data/wrds-bond-returns/>.

Table 5
Portfolio analyses.

Panel A: Well-connected vs. less-connected performance				
	Portfolio return			
	(1)	(2)	(3)	(4)
α	7.798*** (3.522)	7.510*** (3.338)	7.125*** (3.283)	6.966*** (3.208)
Excess market return		0.010 (0.753)	−0.058** (−2.525)	−0.053** (−2.272)
Default premium			0.082*** (3.701)	0.081*** (3.667)
Term spread			0.042*** (2.982)	0.040*** (2.799)
Liquidity factor				−0.008 (−1.184)
Observations	154	154	154	154
R ²		0.004	0.087	0.096
Panel B: Well-connected vs. less-connected performance (by bond risk)				
	Portfolio return			
	High risk		Low risk	
	(1)	(2)	(3)	(4)
α	8.191** (2.169)	9.941*** (2.684)	2.315** (2.221)	2.479** (2.354)
Excess market return		−0.079* (−1.964)		0.020* (1.771)
Default premium		0.007 (0.179)		−0.024** (−2.260)
Term spread		0.012 (0.498)		−0.013* (−1.904)
Liquidity factor		−0.018 (−1.579)		0.003 (1.005)
Observations	154	154	154	154
R ²		0.096		0.041

This table reports the estimates of the monthly portfolio alphas of the buy-minus-sell (e.g., high net buying minus high net selling) portfolios of well-connected and less-connected insurers. In Panel A, we present the alphas and factor loadings for a portfolio that takes a long position in the buy-sell well-connected portfolio and shorts a buy-sell less-connected portfolio (those in the remaining connections quintiles). In Panel B, we present similar analyses within subsamples of bonds in those portfolios based on the bonds' credit and duration risk profiles. The first (second) two columns focus on the subsample of bonds that are both below (above) Category 1-rated securities and have longer (shorter) duration than the median maturity for the sample. Value-weighted portfolios are constructed based on the decile ranks of bonds at the beginning of each month, where the decile ranks are based on buy-sell imbalances from the prior month of trading. For each investor category (i.e., well-connected and less-connected), a buy-sell portfolio is constructed by taking a long (short) position in bonds with the highest net buying (selling), measured as the top (bottom) decile of buy-sell imbalances, by insurers in that connectedness category. Monthly returns for each portfolio are regressed on common bond risk factors: Excess market return (Dickerson et al., 2023), Default premium and Term spread factors (French and Fama, 1989), and the Liquidity factor (Pastor and Stambaugh, 2003). *t*-statistics are in parentheses, and levels of significance are presented as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

(lowest) decile of buy-sell imbalance across bonds traded by two groups: (1) well-connected insurers and (2) less-connected insurers. Similar to our other analyses, we measure each insurer's dealer network based on the previous 12-month trading window in all bonds in the previous month, separating insurers into those in the highest quintile of dealer network size ("well-connected") and all other insurers ("less-connected").²⁰ This long-short portfolio approximates the trading performance by emulating insurers' decision-making in each connectedness group.

In Panel A of Table 5, we present the risk-adjusted abnormal returns (i.e., alphas) of the portfolio that takes a long position in the well-connected insurers' portfolio and a short position in the less-connected insurers' portfolio described above. We base these risk adjustments on commonly used risk models in bond markets: the excess bond-market portfolio as suggested by Dickerson et al. (2023), the term-structure and default bond factors of French and Fama (1989), and the inclusion of the traded (Pastor and Stambaugh, 2003) liquidity factor following Becker and Ivashina (2015). The inclusion of these risk factors in the regressions ensures that portfolio performance is assessed after controlling for differences in risk across the portfolios held by well- and less-connected insurers.

²⁰ To ensure there are sufficient bonds to be formed in each long-short portfolio in each month, the less-connected group aggregates all insurers who are not in the highest quintile. Given the relative illiquidity in these markets, trades are uncommon, particularly for those insurers with fewer dealer connections (which tend to be smaller insurers with less trading activity).

Across all specifications, we find a positive and statistically significant alpha for this long-short portfolio. This highlights that well-connected insurers, on average, outperform less-connected insurers even after adjusting for standard risk factors. We also observe that well-connected insurers' portfolios load more on the 'Default premium' and the 'Term spread,' which suggests that they assume more credit risk (i.e., lower ratings) and term structure risk (i.e., longer maturities).

In our most stringent specification (column 4), the alpha is approximately 7.0 basis points monthly or 0.84% on an annualized basis. Although appearing moderate in economic magnitude, given average monthly returns of approximately 38 basis points across bonds in our sample period (untabulated), this performance still constitutes approximately 18% of outperformance relative to total bond returns in any given month. Also highlighting the economic significance, this advantage totals to around \$15 million in alpha earned by the highest-ranked insurance companies per month, aggregating to \$2.3 billion over the sample period.²¹

We next measure the outperformance of well-connected insurers within subsamples of bonds based on common risk factors. Specifically, we sort bonds by credit and term (i.e., maturity) risk to help hold constant the risk of bonds being compared across portfolios. Moreover, as prior studies show that performance differences are most pronounced in riskier securities (e.g., Gebhardt et al., 2005; Jostova et al., 2013; Painter, 2020), these sorts also allow us to test whether the well-connected insurers' portfolio's outperformance is more pronounced in securities where we would expect credit-specific information to be most important.²²

In Panel B of Table 5, we present subsample analyses based on those presented in Panel A. Columns 1 and 2 present alphas within the high-risk portfolio, which focuses on the subsample of bonds that are both below Category 1-rated securities (i.e., those with NAIC Designation below 1, as seen in Appendix B) and have longer duration than the median maturity for the sample. In column 2, the alpha is approximately 9.9 basis points monthly or 1.20% on an annualized basis. Columns 3 and 4 present alphas within the low-risk portfolio, which focuses on the subsample of bonds that are both above Category 1-rated securities and have shorter duration than the median maturity for the sample. In column 4, the alpha is approximately 2.5 basis points monthly or 0.30% on an annualized basis. The results in this table indicate that our findings are more pronounced for the subsample of bonds for which credit-relevant information is most important for performance (i.e., riskier bonds), further supporting our argument that connections provide an information advantage.

5. Investors' connections to informed dealers

While the results in the prior section provide generalizable evidence that investors with larger dealer networks appear to be more informed, they cannot speak to the source of this information directly. A potential explanation is that larger and more active insurers, who are more likely to transact with extensive dealer networks, also possess superior internal resources, such as higher-quality investment teams.

In this section, we address this endogeneity concern by isolating information transmitted through dealer networks rather than investors' internal capabilities. To do so, we study the individual connections themselves in two narrower settings where particular dealers are likely to be informed about specific events: downgrades and M&As. Using these settings, we can observe how information is expected to flow along specific trading connections, thereby strengthening our inferences regarding the likely source of the information.

5.1. Credit rating downgrades

Motivated by our findings in Section 4, we first focus on insurers' trades around credit rating downgrades. We focus on instances where connections to certain, informed dealers are more likely than others to yield private information about upcoming downgrades. Based on prior evidence, we focus on two instances where certain dealers are more likely to be informed about particular downgrades. First, we expect that dealers who handled the bond issuance are much more likely than other dealers to have an information advantage in that particular security (e.g., Cliff and Denis, 2004; De Franco et al., 2009; Chiang et al., 2019). Second, dealers who tend to be primary market-makers in a bond issue are likely to be highly informed about that particular security (e.g., Babus and Kondor, 2018; Hagströmer and Menkveld, 2019).

We use the above proxies in an event-study design on the sample of downgrades described in Section 3.3. Specifically, we assess whether the trading clients of more informed dealers are more likely to trade in an informed manner in the short window (30 trading days) before the downgrade. Specifically, we estimate the following regression:

$$\begin{aligned} \text{Insurer } BSI_{i,e} = & \beta_1 \text{InformedConnection}_{i,e} + \beta_2 \text{InformedConnection}_{i,e} \times \text{to Junk}_e \\ & + \Sigma \beta_e \text{Event}_e + \Sigma \beta_i \text{Insurer}_i + \epsilon_{i,e}, \end{aligned} \quad (2)$$

where $\text{Insurer } BSI_{i,e}$ is the insurer i 's pre-downgrade buy-sell imbalance in corporate bonds for the firm with downgrade event e . $\text{InformedConnection}_{i,e}$ is one of several variables denoting a trading connection, or the strength of a trading connection, between the informed dealer and insurer i in the 12-months before downgrade event e . to Junk_e , a variable that indicates whether the focal

²¹ Using a back-of-the-envelope calculation, the average monthly volume traded by the well-connected insurers is \$21.9 billion. Assuming trading positions are held for a month at a time, these insurers earn a total of \$21.9 billion $\times$ 7 basis points $\approx$ \$15 million per month. This aggregates to \$15 million $\times$ 154 months in our sample $\approx$ \$2.3 billion.

²² Specifically, this finding supports our hypotheses, in that these securities tend to be the most sensitive to credit-specific information (e.g., Even-Tov, 2017; Huber et al., 2024).

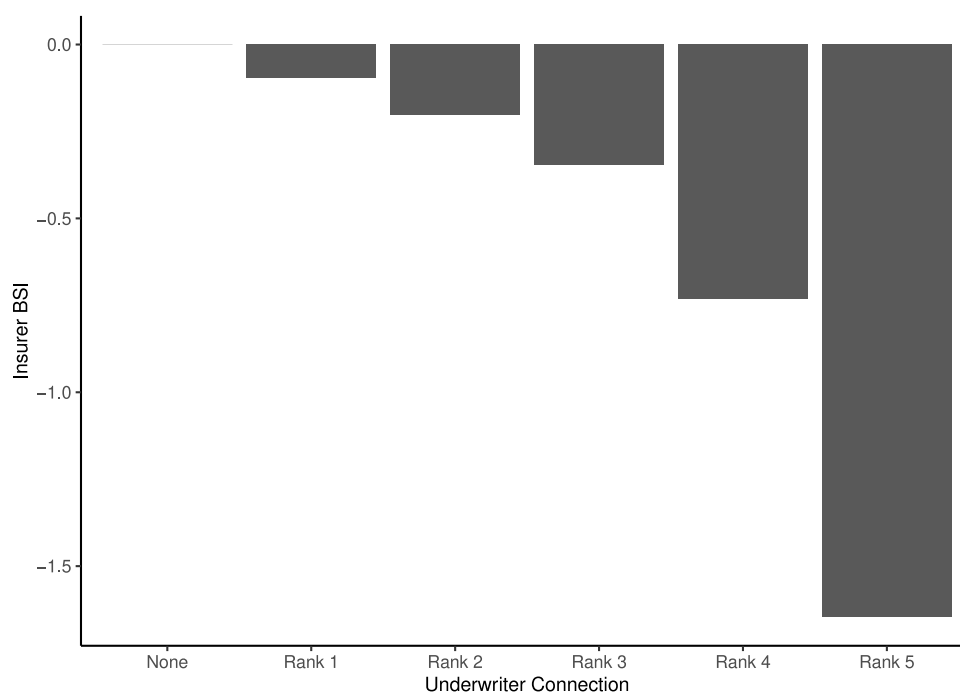


Fig. 3. Insurers' connections and buy-sell imbalances in bonds before junk downgrades.

This figure presents the average of *Insurer BSI* bonds over the $[-30, -1]$ window of a junk downgrade based on the rank of the strength of the insurers' trading connections with the dealers affiliated with the bond's underwriter. Insurers in the underwriter connection category of "None" do not have any trading activity with the dealer affiliated with the underwriter in the 12-month window before the announcement. Insurers in the underwriter connection category of "Rank K " have nonzero trading activity with the dealer affiliated with the underwriter in the 12-month window before the junk downgrade that is in the K th quintile of that dealer's insurer clients within the same window.

bond has been downgraded from investment-grade to high-yield, allows us to explore decision-making around credit rating changes that insurers have the strongest incentives to avoid (i.e., see Section 4.1).

We include event fixed effects, which hold constant other factors with limited variation in the short window around the event (i.e., time-varying controls such as time to maturity). We also include insurer fixed effects, which subsume any time-invariant insurance company characteristics that could confound our interpretation. For example, they help us rule out the alternative explanation that some insurers are better at trading prior to downgrades because they consistently have better in-house resources.

Table 6 presents regression results for multiple variations of Eq. (2). Across all specifications, we find evidence that private information about forthcoming downgrades flows from informed dealers to their trading clients. Specifically, the coefficients on β_1 and β_2 are negative and statistically significant at the 1% level across specifications. Our estimates in column 1 of both panels indicate that, when an insurer has a trading connection with a dealer who issued the bond or is a primary market-maker for the bond, that insurer is more likely to sell that bond ahead of a downgrade. This result is particularly pronounced ahead of downgrades to junk status, as we would expect given insurers' incentives. In columns 2 and 3 of both panels, we also find that these results hold when using measures of the strength of insurers' connections to informed dealers.

To emphasize these relationships, Fig. 3 illustrates these findings in the context of junk downgrades, the most powerful setting to observe these behaviors. We see a clear relation between the importance of the insurer to the dealer who underwrote the downgraded bond and the insurer's propensity to sell bonds ahead of junk downgrades. In the case of no connection, we see neutral buy-sell activity (i.e., no predictability).

5.2. Mergers & Acquisitions

We add to the above evidence by focusing on a setting prior research commonly uses to explore the presence of informed trading—M&A events (e.g., Bhattacharya, 2014; Kacperczyk and Pagnotta, 2019; Augustin and Subrahmanyam, 2020; Suk and Wang, 2021). Although rare compared to many of the other types of information insurers are likely to learn through their dealer networks, M&A events provide an ideal setting to explore dealer-insurer information flows for several reasons. First, the information is non-public but known to the specific dealer-affiliated banks involved in the transaction (e.g., Acharya and Johnson, 2010; Kang, 2024),

Table 6
Investor connections and bond downgrades.

Panel A: Underwriter connections			
	Insurer BSI		
	(1)	(2)	(3)
Underwriter Connection	−0.036*** (−10.025)		
Underwriter Connection × to Junk	−0.942*** (−15.720)		
Underwriter Connection Rank		−0.027*** (−13.400)	
Underwriter Connection Rank × to Junk		−0.386*** (−17.631)	
Underwriter Connection Strength			−0.207*** (−7.054)
Underwriter Connection Strength × to Junk			−3.618*** (−17.643)
Event fixed effects	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes
Observations	87,655,544	87,655,544	87,655,544
R ²	0.051	0.051	0.051
Panel B: Specialist connections			
	Insurer BSI		
	(1)	(2)	(3)
Specialist Connection	−0.376*** (−18.344)		
Specialist Connection × to Junk	−0.980*** (−15.066)		
Specialist Connection Rank		−0.166*** (−19.125)	
Specialist Connection Rank × to Junk		−0.418*** (−17.029)	
Specialist Connection Strength			−0.12713*** (−4.522)
Specialist Connection Strength × to Junk			−0.76338*** (−5.845)
Event fixed effects	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes
Observations	87,655,544	87,655,544	87,655,544
R ²	0.051	0.052	0.051

This table presents analyses that examine the relation between existing trading connections and an insurer's buy-sell imbalance in the downgraded company's bonds during the 30 trading day period before a downgrade. Panel A examines insurers' connection to the dealers affiliated with banks that underwrote a bond issue. Panel B examines insurers' connection to the dealers that are specialist market makers in the downgraded bond. We estimate several versions of Eq. (2) using the sample of insurer-event observations described in Section 3. All variables are defined in Appendix A. *t*-statistics based on standard errors clustered by insurer are in parentheses. Levels of significance are presented as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

and therefore, those affiliated dealers are likely to be the primary source of this information for insurers.²³ Second, M&As can have important effects on bond prices and credit ratings (e.g., Asquith and Wizman, 1990; Bodnaruk and Rossi, 2016; Eisenthal-Berkovitz et al., 2020), so insurers have powerful incentives to trade on this information.²⁴

Again leveraging an event-study design on the sample of M&As we describe in Section 3.3, we assess whether the trading clients of deal-affiliated dealers are more likely to trade in an informed manner in the short window (30 trading days) before the deal

²³ These findings are indicative of the leakage of MNPI before upcoming M&A announcements. We note, however, that there are also often “rumors” of these deals (e.g., Alperovych et al., 2021). For instance, deal rumors related to the Dell (Damouni and Gupta, 2013; Toll-Reed and Schrank, 2013) and Staples (Hirsch and Ramakrishnan, 2017; Dye, 2017) leveraged buyouts occurred more than four months before the official deal announcements, which were followed by subsequent downgrades to junk status due to the increased debt associated with the deal. Our findings could also indicate that connected investors are better able to discern whether a rumor is unfounded or not, perhaps by combining multiple pieces of immaterial information into a “mosaic” (e.g., Solomon and Soltes, 2015; Augustin et al., 2019). Consistent with this, the enforcement, and prosecution, of insider trading around MNPI leaks is difficult in practice.

²⁴ For example, 76.5% of M&A deals we study are accompanied by negative returns, and 2.2% are accompanied by a downgrade for the target company's bonds within three months of the deal announcement. The downgrade probability increases to 19% in the 12 months after the deal announcement, during which time the terms of the deal might become effective.

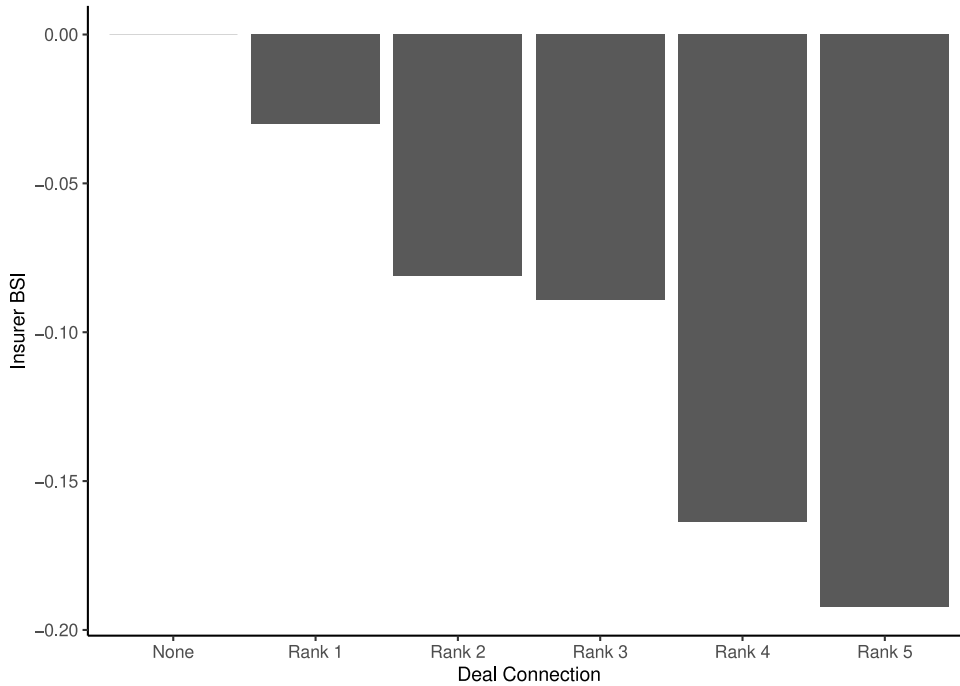


Fig. 4. Insurers' connections and buy-sell imbalances in bonds before adverse M&A events.

This figure presents the average of *Insurer BSI* in M&A target bonds over the $[-30, -1]$ window of deal announcements based on the rank of the strength of the insurers' trading connections with deal-affiliated dealers. It presents the BSI for M&As with negative announcement returns in the target bonds. Insurers in the deal connection category of "None" do not have any trading activity with the deal-affiliated dealer in the 12-month window before the announcement. Insurers in the deal connection category of "Rank K " have nonzero trading activity with the deal-affiliated dealer in the 12-month window before the M&A announcement that is in the K th quintile of that dealer's insurer clients within the same window.

is publicly announced. Following Haselmann et al. (2023), we use non-adverse events as a baseline to which to compare adverse events and estimate the following regression:

$$\begin{aligned} \text{Insurer } BSI_{i,e} = & \beta_1 \text{DealConnection}_{i,e} + \beta_2 \text{DealConnection}_{i,e} \times \text{NegativeEvent}_e \\ & + \Sigma \beta_e \text{Event}_e + \Sigma \beta_i \text{Insurer}_i + \epsilon_{i,e}, \end{aligned} \quad (3)$$

where *Insurer BSI_{i,e}* is the insurer i 's pre-event buy-sell imbalance in corporate bonds for the firm that is the target of the deal event e and *DealConnection_{i,e}* is one of several variables denoting the number and strength of trading connections between the deal-affiliated dealer and insurer i for the deal event e . *NegativeEvent* is one of several variables indicating whether or not the M&A deal is a negative event for the target company's bonds. Our fixed effects structure follows the same logic as outlined in the prior section.

Table 7 presents regression results for multiple variations of Eq. (3). Across all specifications, we find evidence that private information about forthcoming M&A deals flows from deal-affiliated dealers to their trading clients. In column 1 of both panels, the coefficient β_2 is negative and statistically significant at the 1% level, indicating insurers with a trading connection with a dealer who is affiliated with an M&A deal tend to sell bonds to a greater extent ahead of M&A announcements accompanied by negative value implications. Additionally, column 2 highlights that the results are more pronounced for insurers with a greater number of unique connections to the deal, as we would expect if insurers are sourcing this information from their dealer network.

In columns 3 and 4, we examine the strength of the connection between the insurer and dealer. Our evidence is consistent with deal-affiliated dealers providing information to their best trading clients. Specifically, we find a negative and statistically significant (at the 5% level or lower threshold) coefficient β_2 . This indicates that the prevalence of informed trade by insurers before an M&A announcement with adverse credit consequences increases with past order flow between the insurer and deal-affiliated dealer.

Fig. 4 displays the above findings graphically. We see a clear relation between the importance of the insurer to the dealer, based on the rank of the insurer's trading flow as a proportion of the dealer's total trading flow with insurers, and the insurer's propensity to sell target bonds before M&As with negative announcement returns are publicly announced. Consistent with the above findings, in the case of no connection, we see neutral buy-sell activity.

Table 7
Investors' connections to deal-affiliated dealers around M&A activity.

Panel A: Post-deal downgrades				
	Insurer BSI			
	(1)	(2)	(3)	(4)
Deal Connection $\times$ Downgrade ^{3mo.}	−0.456*** (−4.543)			
# Deal Connections $\times$ Downgrade ^{3mo.}		−0.218*** (−3.846)		
Deal Connection Strength $\times$ Downgrade ^{3mo.}			−0.890** (−2.020)	
Deal Connection Rank $\times$ Downgrade ^{3mo.}				−0.141*** (−3.361)
Main effects included	Yes	Yes	Yes	Yes
Event fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes
Observations	854,789	854,789	854,789	854,789
R ²	0.011	0.011	0.011	0.011
Panel B: Negative deal announcement returns				
	Insurer BSI			
	(1)	(2)	(3)	(4)
Deal Connection $\times$ Negative Return	−0.261*** (−5.654)			
# Deal Connections $\times$ Negative Return		−0.131*** (−5.210)		
Deal Connection Strength $\times$ Negative Return			−0.451** (−2.204)	
Deal Connection Rank $\times$ Negative Return				−0.090*** (−4.726)
Main effects included	Yes	Yes	Yes	Yes
Event fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes
Observations	486,044	486,044	486,044	486,044
R ²	0.013	0.013	0.013	0.013

This table presents analyses that examine the relation between existing trading connections to the dealers affiliated with banks that are advisors on an M&A deal and the insurer's buy-sell imbalance in the target company bonds during the 30 trading day period before the announcement of a takeover. We estimate several versions of Eq. (3) using the sample of insurer-event observations described in Section 3. Panel A presents results that estimate the differences in trading patterns of deal-connected insurers for M&A events with and without post-deal target company bond downgrades. Panel B presents results that estimate the differences in trading patterns of deal-connected insurers for M&A events with and without negative deal returns. All variables are defined in Appendix A. *t*-statistics based on standard errors clustered by insurer are in parentheses. Levels of significance are presented as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

6. Additional analyses

6.1. Dealer incentives: Payment for future order flow

Our evidence thus far indicates that a key incentive for dealers to provide information to investors is to reward them for trading flow. This type of soft dollar arrangement, whereby information is exchanged for trading flow, is consistent with prior studies' findings in other settings (e.g., Irvine, 2000; Frankel et al., 2006; Green et al., 2014), as well as anecdotal conversations with practitioners (e.g., see Section 2). While our main analyses focus on past trading flow as an indicator of the strength of these relationships, an important question is whether such information exchange leads to future commissions for the dealer, similar to what related studies find (e.g., Green et al., 2014). To provide evidence of this “quid pro quo” mechanism, we take two approaches using the M&A setting, which provides a powerful setting to study the link between the likely source of private information and investors' trading connections.

We first investigate whether there is a relation between future trading flow and informed trading around M&A deals. Specifically, we estimate versions of Eq. (3) in which we replace the past trading *DealConnection* with variables denoting future trading relationships (i.e., in the next 12 months) between the insurer and the deal-affiliated dealer. We present these analyses in Panel A of Table 8. Across all specifications, we find that the coefficient β_2 is negative and statistically significant at the 1% level, indicating that selling is more likely to occur before adverse M&A events when an insurer provides trading flow to a deal-affiliated dealer *after* the event.

Table 8

Dealer incentives: Information for future trading flow.

Panel A: Future trading connections				
	Insurer BSI			
	(1)	(2)	(3)	(4)
Future Deal Connection $\times$ Downgrade ^{3mo.}	−0.448*** (−4.602)			
Future Deal Connection Rank $\times$ Downgrade ^{3mo.}		−0.132*** (−3.247)		
Future Deal Connection $\times$ Negative Return			−0.220*** (−4.837)	
Future Deal Connection Rank $\times$ Negative Return				−0.072*** (−3.870)
Main effects included	Yes	Yes	Yes	Yes
Event fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes
Observations	910,369	910,369	522,552	522,552
R ²	0.010	0.010	0.012	0.012
Panel B: Changes in trading flow				
	Δ Trade Share Deal Connection			
	(1)	(2)	(3)	(4)
BSI < 0 $\times$ Downgrade ^{3mo.}	0.070** (2.021)	0.075*** (2.599)		
BSI < 0 $\times$ Negative Return			0.042* (1.793)	0.048** (2.353)
BSI < 0	−0.078*** (−6.428)	0.087*** (−9.702)	−0.109*** (−4.798)	0.030* (1.67)
Event fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	No	Yes	No	Yes
Observations	838,432	838,432	477,528	477,528
R ²	0.06	0.414	0.062	0.413

This table presents additional analyses that examine potential dealer benefits. Panel A examines the relation between the strength of insurers' future trading connection to the dealers affiliated with banks that are advisors on an M&A deal and the insurer's buy-sell imbalance in the target company bonds during the 30 trading day period before the announcement of a takeover. We estimate several versions of Eq. (3) using the sample of insurer-event observations described in Section 3. Panel B presents the results examining investors' dealer choices. The dependent variable Δ Trade Share Deal Connection is the change in the share of an investor's trading volume executed with dealers connected to the M&A transaction, measured as the difference between the 12-month post-deal and 12-month pre-deal periods. Insurer BSI < 0 is an indicator variable equal to one if the insurer had a net sell position in the 30 trading days ahead of the takeover announcement. All variables are also defined in Appendix A. *t*-statistics based on standard errors clustered by insurer are in parentheses. Levels of significance are presented as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Second, we explore whether insurers allocate a larger share of their future trading flow to dealers whose private information previously benefited those insurers. Specifically, we test whether insurers with pre-M&A sell decisions that avoid downgrades or negative returns increase their trading flows with deal-affiliated dealers to a greater extent than other insurers after the M&A event. Using the same sample as above, we estimate versions of the following regression:

$$\begin{aligned}
 \Delta \text{ Trade Share Deal Connection}_{i,e} = & \beta_1 \text{BSI} < 0_{i,e} \\
 & + \beta_2 \text{BSI} < 0_{i,e} \times \text{NegativeEvent}_e \\
 & + \Sigma \beta_e \text{Event}_e + \Sigma \beta_i \text{Insurer}_i + \epsilon_{i,e},
 \end{aligned} \tag{4}$$

where Δ Trade Share Deal Connection is the change in the share of insurer i 's trading volume allocated to the deal-affiliated dealer, between the 12 months prior to the event e and the 12 months after the event. $\text{BSI} < 0_{i,e}$ is an indicator variable for whether the insurer i 's pre-event buy-sell imbalance in corporate bonds for the firm that is the target of the deal event e is negative. *NegativeEvent* is one of several variables indicating whether or not the M&A deal is a negative event for the target company's bonds. In the most stringent specification, the fixed effects structure follows Eq. (3) and includes event and insurer fixed effects.

To the extent that dealers' trading clients pay for information with future order flow, we expect β_2 to be positive, indicating that insurers who are net sellers prior to the adverse deal announcement increase their trading flow with the deal-affiliated dealer to a greater extent after the event, compared to other insurers. Our results, presented in Panel B of Table 8, support this possibility. Specifically, across all specifications, the coefficient β_2 is positive and significant. Insurers appear to direct more trading flow to deal-affiliated dealers when their pre-M&A trades avoid losses. These findings provide compelling evidence of the incentives for

Table 9
Unsophisticated counterparties and informed trading.

Panel A: Retail trading following insurer sell transactions			
	Retail BSI ^{Week0} (1)	Retail BSI ^{Week1} (2)	Retail BSI ^{Week2} (3)
Sell	0.106*** (33.411)	0.060*** (29.384)	0.053*** (27.083)
Controls	Yes	Yes	Yes
Rating fixed effects	Yes	Yes	Yes
Year × Month fixed effects	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes
Observations	1,769,604	1,769,604	1,769,604
R ²	0.262	0.253	0.246
Panel B: Retail BSI ahead of M&A events			
	Retail BSI		
	(1)	(2)	
Downgrade ^{3mo.}	0.019*** (8.757)		
Negative Return		0.038*** (6.283)	
Event time fixed effects	Yes	Yes	
Observations	16,200	9210	
R ²	0.002	0.005	

This table presents analyses of retail trading behavior around the trades and events of the main tables. Panel A examines weekly Retail BSI measured in calendar weeks 0, 1, and 2 after an insurer's trade in a particular bond. *t*-statistics based on standard errors clustered by insurer are in parentheses. Panel B examines daily Retail BSI in the 30 trading days ahead of the M&A events of the analyses in Table 7. *t*-statistics based on standard errors clustered by event-day are in parentheses. Levels of significance are presented as follows: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

dealers to provide investors with private information as part of a “quid pro quo”, not just to maintain but also to actually strengthen those relationships and gain future trading commissions.²⁵

6.2. Unsophisticated investors' trading around insurers' trades

An important question is who loses due to the well-connected insurers' information advantages we document. Although data limitations prevent us from investigating all investors' losses, as we only observe a small portion of trades in our sample, we provide evidence on these issues among arguably the most important subset of less sophisticated investors—retail investors. While retail investors are unlikely to be the only counterparties absorbing informed insurer trades, retail investor volume represents a substantial share — approximately 40% — of insurer trading volume (\$100 bn compared to \$250bn annually). We also focus on retail investor trades because they often, but not always, reflect activity by less-informed investors, and are the subject of regulatory interest (e.g., SEC, 2014, 2018) and extensive prior research (e.g., Cuny, 2018; Cuny et al., 2021; deHaan et al., 2023).

We use TRACE data and proxy for retail bond trades as those \$100,000 or less, in line with standard practice (e.g., Larcker and Watts, 2020; O'Hara and Zhou, 2021; Huber et al., 2024).²⁶ Based on these data, we measure retail investors' buy-sell imbalances around insurer trades. To the extent that insurers are trading on their information advantages at the expense of less sophisticated investors, we expect retail investors to be net buyers when insurers are selling.

We present two separate analyses in Table 9 to provide evidence of these behaviors. In Panel A, we use the sample of insurers' transactions described in Section 3 to explore whether retail investors are, on average, on the other side of insurers' trades. Specifically, we assess whether retail investors are more likely to be net buyers of securities in the period around, and immediately after, insurers' sell transactions.²⁷ Our evidence shows a strong negative relation between insurers' trades and retail investor net

²⁵ In addition to the analyses presented in this section, we note two additional findings. First, we highlight trading relationships tend to be very stable over time, with an autocorrelation of 68.7%. This suggests that much of the information provided is intended to build these longstanding relationships, with the information potentially serving as payment for past order flow. Second, our findings in Section 4 are qualitatively unchanged when using *future* as opposed to *past* dealer networks.

²⁶ Retail-sized trades are reliably identifiable in largely anonymous transaction data because, unlike in equity markets, trade size remains a reliable proxy for distinguishing between retail and institutional trades, because trade splitting is infrequent (i.e., see Section 6.3). Therefore, practitioners and regulators widely attribute trades below \$100,000 to retail investors. For a detailed description, see deHaan et al. (2023).

²⁷ When providing liquidity, dealers typically buy bonds into their own inventory, although they cannot hold these risks for long periods (e.g., Goldstein and Hotchkiss, 2020). Using multiple measurement windows accounts for the fact that it takes time for dealers to work off larger positions, such as those that insurers typically trade.

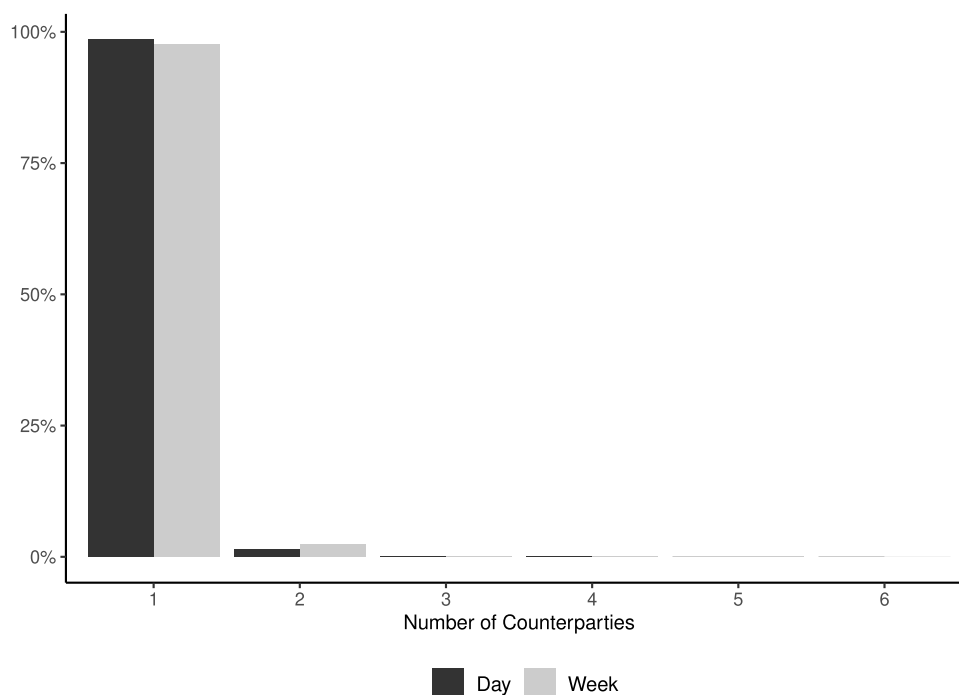


Fig. 5. Number of trading counterparties.

This figure shows the prevalence of trade splitting by insurers by showing how many counterparties insurance companies transact with in the same bond on the same day (black bars) and week (gray bars).

demand, implying that retail investors tend to be the ultimate counterparties to insurers' transactions. In economic terms, an insurer's sale coincides with a 10 percentage point increase in retail investors' net buying activities. The level of retail buying activity stays elevated by 6 and 5 percentage points in the first and second weeks after the sale, respectively.

In Panel B, we study whether retail investors tend to be net buyers of securities before adverse M&A events, based on our prior evidence in Section 5.2 showing that insurers connected to deal-affiliated dealers tend to be the selling counterparties. Specifically, using the same sample of events, we explore the dynamics of retail buy-sell imbalances in the 30 days prior to M&A transactions. As before, we use non-adverse events as a baseline, and explore whether retail investors are more likely to be net buyers when the M&A event has negative implications for the firm's credit, as measured by post-deal downgrades (column 1) or negative announcement returns (column 2).

In both columns, the coefficient on the negative event indicator is positive and statistically significant at the 1% level. This indicates that retail investors are more likely to be net buyers before negative M&A events, precisely the events before which insurers with connections to deal-affiliated dealers are selling. Combined with our results reported in Panel A, this finding highlights that less sophisticated retail investors are one party that tends to be harmed by insurers who obtain information advantages through their dealer networks. Together, these results highlight a new channel through which dealers exploit retail investors, adding to prior evidence that primarily focuses on disadvantages in the realm of liquidity and execution costs (e.g., [Cuny, 2018](#); [Cuny et al., 2021](#); [Griffin et al., 2023](#)).

6.3. Alternative explanations

Our findings indicate that insurers with larger dealer networks tend to make more informed trading decisions, and that these performance advantages are obtained from information that passes through insurers' trading networks. Based on prior studies, we consider several of the plausible alternative explanations for our findings.

To begin, prior studies highlight that informed traders split their trades across multiple dealers to hide their information (e.g., such as in Treasury markets, as shown in [Kondor and Pintér, 2022](#)), effectively expanding their trading networks when they are more informed. If trade splitting occurs in our setting, it could explain our results. To address this possibility, we consider the propensity with which we observe trade splitting in our sample. Specifically, we compute the number of counterparties that insurers transact with when trading a particular bond. [Fig. 5](#) shows that, in more than 95% of trades, the insurer transacts with only one dealer counterparty when trading a given bond within a day. We see similar results even when extending the trading period to the

Table 10

Alternative explanation: Feedback between bond illiquidity and adverse credit events.

Panel A: Trade size fixed effects				
	Junk Downgrade ^{3mo.}		Default ^{1Y}	
	(1)	(2)	(3)	(4)
Sell × Connection Rank	0.037*** (4.217)	0.040*** (4.572)	0.026*** (3.674)	0.024*** (4.767)
Connection Rank	−0.041*** (−5.946)	−0.023*** (−3.010)	−0.017*** (−3.037)	−0.004 (−0.754)
Sell	0.028 (0.821)	−0.054 (−1.474)	−0.002 (−0.053)	−0.073*** (−3.436)
Controls	Yes	Yes	Yes	Yes
Rating fixed effects	Yes	Yes	Yes	Yes
Trade Size fixed effects	Yes	Yes	Yes	Yes
Year × Month fixed effects	No	Yes	No	Yes
Insurer fixed effects	No	Yes	No	Yes
Bond fixed effects	No	Yes	No	Yes
Observations	1,540,829	1,540,829	1,560,073	1,560,073
R ²	0.032	0.172	0.132	0.662
Panel B: Cross-sectional variation in bond liquidity				
Dealer Volume	Junk Downgrade ^{3mo.}		Default ^{1Y}	
	High (1)	Low (2)	High (3)	Low (4)
Sell × Connection Rank	0.046*** (3.437)	0.009 (0.882)	0.035*** (4.687)	0.006 (1.006)
Connection Rank	−0.030*** (−3.228)	−0.004 (−0.393)	−0.010 (−1.515)	0.002 (0.331)
Sell	−0.081 (−1.442)	0.025 (0.555)	−0.134*** (−4.042)	−0.035 (−1.253)
Controls	Yes	Yes	Yes	Yes
Rating fixed effects	Yes	Yes	Yes	Yes
Trade Size fixed effects	Yes	Yes	Yes	Yes
Year × Month fixed effects	Yes	Yes	Yes	Yes
Insurer fixed effects	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	Yes
Observations	562,998	681,146	566,336	692,586
R ²	0.251	0.183	0.739	0.704

This table presents results from robustness checks showing that the adverse credit event results are not driven by insurance company trades in illiquid bond markets causing the events. Panel A adds trade size decile fixed effects to the transaction-level analyses in Panel B of [Table 3](#) to control for the liquidity demand by insurance companies. Panel B shows results for the subsamples of bonds with above or below monthly median level of dealer intermediation. All variables are defined in [Appendix A](#). *t*-statistics based on standard errors clustered by insurer are in parentheses. Levels of significance are presented as follows:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

same week. These statistics are nearly identical when using the dollar value of trades rather than the number of trades (untabulated). Therefore, we conclude that trade splitting cannot explain our findings.²⁸

Additionally, [He and Milbradt \(2014\)](#) highlights that investors may cause, rather than predict, adverse credit events by worsening the illiquidity of bonds through their trading (i.e., “default-liquidity spirals”). To the extent this occurs, well-connected insurers (who tend to be larger) would be selling into an illiquid secondary market, exacerbating downward price pressure. This would cause other market participants and rating agencies to interpret these sales as signals of deteriorating credit fundamentals, thereby accelerating credit downgrades or defaults. We rule out this possibility in two ways. First, we include trade size fixed effects, which help to account for the effect of liquidity demand. Panel A of [Table 10](#) shows that, even controlling for the size of the trade (i.e., the required depth), our results remain effectively unchanged. Second, we split our sample into high- and low-liquidity bonds based on median dealer volume, to investigate whether performance differences are more pronounced in the least liquid bonds (i.e., where there is little liquidity supply). Panel B of [Table 10](#) shows that the performance differences based on investors’ networks are concentrated in the most liquid bonds, which is the opposite of what we would expect to find if default-liquidity spirals were occurring. One possibility for this finding is that informed traders target pockets with little market impact, on average, similar to what prior research finds in equity markets (e.g., [Collin-Dufresne and Fos, 2015](#); [Kacperczyk and Pagnotta, 2019](#)).

²⁸ This lack of trade splitting can be explained by the institutional features of the corporate bond market. Algorithmic trading, and thus trade splitting, is limited during the period we study (e.g., [O’Hara and Zhou, 2021](#)). Additionally, large investors often have little incentive to split their trades, because they do not typically trade anonymously and transaction costs tend to be higher for smaller trades (e.g., [Bessembinder et al., 2020](#)).

7. Concluding remarks

Our study explores the informational value of trading networks in OTC markets. Leveraging novel regulatory data allowing us to observe the entire trading network of U.S. insurance companies in corporate bond markets, we provide evidence that insurers gather private information about corporate fundamentals through their dealer networks. Thus, we show that information is an important benefit provided through dealer networks, which helps explain why, in practice, these networks appear larger than would be optimal if liquidity were the only consideration.

We emphasize that an insurer's decision to develop an extensive dealer network and acquire private information is endogenous. Although our empirical analyses and anecdotal evidence support the inference that insurers obtain information through their dealer networks, we acknowledge that some of our estimates capture this joint decision by insurers. However, our collective analyses suggest that a "quid pro quo" mechanism of information for trading flow plays a key role in the formation and maintenance of network connections.

While many of these information flows are likely legal, our findings suggest they may provide unfair advantages to dealers' top trading clients. This is particularly true of several of our findings that point to MNPI leakages at broker-dealers. Given this, our findings potentially also offer valuable insights to regulators, who have historically devoted fewer enforcement resources to these markets than to the equity market.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Variable definitions

This table contains descriptions of the primary variables used throughout this paper. Sources include S&P Capital IQ (S&P), Mergent FISD (Mergent), Bond Returns by WRDS (WRDS Bond), Fama–French portfolios and factors on WRDS (WRDS FF), [Dickerson et al. \(2023\)](#) data, SDC Platinum (SDC), TRACE transactions (TRACE), and IBES. All continuous variables, except returns, are winsorized at the 1st and 99th percentiles to mitigate the effect of outliers.

Variable	Definition
Transaction-level analyses	
AUM	The natural log of the par value of all corporate bonds held by the insurance company as of the fiscal year end, in \$ billions. (S&P)
Bond Size	The natural log of the bond's offering amount, in \$ billions. (Mergent)
Connection Rank	Quintile rank (0 = Low, 4 = High) of the number of dealers with whom the insurer transacts in the 12-month window prior to the transaction. (S&P)
Dealer Volume	Bond-level monthly par value traded in interdealer transactions. (TRACE)
Default ^{1Y}	Indicator variable set to 100 if the bond's issuer defaults over the next year. (Mergent)
Downgrade ^h	Indicator variable set to 100 if the bond experienced a downgrade by S&P over the next <i>h</i> months. Superscript <i>h</i> indicates the measurement horizon: 3 months and 12 months. (Mergent)
Junk Downgrade ^{3mo.}	Indicator variable set to 100 if the bond experienced a downgrade to junk status, defined as a downgrade from BB- or higher to B+ or lower, by S&P over the next 3 months. (Mergent)
M&A Connection Rank	Quintile rank (0 = Low, 4 = High) of the number of M&A advisor-affiliated dealers with whom the insurer transacts in the 12-month window prior to the transaction. M&A advisor-affiliated dealers are those whose investment banking division advised at least one M&A deal in the previous 12 months. (S&P, SDC)
Non-M&A Connection Rank	Quintile rank (0 = Low, 4 = High) of the number of non-M&A advisor-affiliated dealers with whom the insurer transacts in the 12-month window prior to the transaction. (S&P, SDC)
Non-Research Connection Rank	Quintile rank (0 = Low, 4 = High) of the number of non-research-affiliated dealers with whom the insurer transacts in the 12-month window prior to the transaction. (S&P, IBES)
Rating	Credit rating level for the bond, where Rating = 1 denotes a AAA rating and Rating = 22 denotes a D rating. The WRDS Bond Returns data assigns the value of S&P ratings, if available, Moody's if S&P is missing, and Fitch if both S&P and Moody's are missing. (Mergent)
Research Connection Rank	Quintile rank (0 = Low, 4 = High) of the number of research-affiliated dealers with whom the insurer transacts in the 12-month window prior to the transaction. (S&P, IBES)
Retail BSI ^{Weekh}	Weekly buy-sell imbalance of retail-sized trades (<\$100k par volume) in a bond <i>h</i> calendar weeks after the insurer's trade in the bond. (TRACE)
Sell	Indicator variable set to 1 if the transaction is a sell transaction for the insurer. (S&P)

Time to Maturity	The bond's remaining time to maturity, in years. (Mergent)
Yearly Trade Volume	The natural log of an insurance company's total par value traded during the calendar year, in \$ billions. (S&P)
Monthly portfolio-level analyses	
Default premium	Return difference between the market portfolio of long-term corporate bonds and long-term government bonds (French and Fama, 1989; Dickerson et al., 2023)
Excess market return	Average return on all bonds, weighted by their amounts outstanding (Dickerson et al., 2023).
Liquidity factor	Traded liquidity factor (Pastor and Stambaugh, 2003). (WRDS FF)
Portfolio return	Monthly portfolio returns of a portfolio that takes a long position in the buy-sell well-connected insurers' portfolio and shorts a buy-sell less well-connected insurers' portfolio, measured in basis points. (S&P, WRDS Bond)
Term spread	Return difference between long-term government bonds and the one-month T-Bill rate (French and Fama, 1989). (Dickerson et al., 2023)
Insurer-event level analyses	
# Deal Connections	Number of deal-affiliated dealers with whom the insurer transacted in the 12-month window prior to the event. Deal-affiliated dealers are those affiliated with deal advisors of either the target or acquirer. (S&P, SDC)
Δ Trade Share Deal Connection	Change in the fraction of an insurer's par value trading volume executed with deal-affiliated dealers between the 12 months before and the 12 months following an M&A announcement. (S&P, SDC)
BSI < 0	Indicator variable set to 1 if Insurer BSI was less than 0 ahead of M&A announcement. (S&P)
Deal Connection	Indicator variable set to 1 if the insurer transacted with a deal-affiliated dealer in the 12-month window prior to the event. Deal-affiliated dealers are those affiliated with deal advisors of either the target or acquirer. (S&P, SDC)
Deal Connection Rank	Quintile rank (0 = None, 1 = Low, 5 = High) of the proportion of the deal-affiliated dealer's transaction volume that is with the insurer in the 12-month window prior to the event, measured in percentage points. We determine the rank of this proportion within the dealer's insurer clients with nonzero trading volume with that dealer within the same window. Then, we take the maximum rank across all deal-affiliated dealers, where deal-affiliated dealers are those affiliated with deal advisors of either the target or acquirer. Deal Connection Rank is set to zero if Deal Connection Strength = 0. (S&P, SDC)
Deal Connection Strength	Proportion of the deal-affiliated dealer's transaction volume that is with the insurer in the 12-month window prior to the event, measured in percentage points. This proportion is the insurer-year's maximum across all deal-affiliated dealers, where deal-affiliated dealers are those affiliated with deal advisors of either the target or acquirer. (S&P, SDC)
Downgrade ^{3mo.}	Indicator variable set to 1 if any of the target company's bonds experienced a downgrade by S&P over the next 3 months after the deal announcement. (Mergent)
Insurer BSI	The aggregate buy-sell imbalance of the insurer in the 30 trading days prior to the deal announcement, measured in percentage points. The buy-sell imbalance is calculated as the difference between the insurer's buy volume and sell volume, scaled by the sum of the buy and sell volume and multiplied by 100. (S&P)
Future Deal Connection	<i>Deal Connection</i> measured over the 12-month window following the deal announcement month. (S&P, SDC)
Future Deal Connection Rank	<i>Deal Connection Rank</i> measured over the 12-month window following the deal announcement month. (S&P, SDC)
Negative Return	Indicator variable set to 1 if the weighted average deal announcement return for the target company's bonds is negative. For each outstanding bond, we calculate its deal announcement return as the percentage difference between the earliest price available in the 10 trading days before the deal announcement and the earliest price available in the 10 trading days after the deal announcement. The average deal announcement return is the weighted average across all outstanding target company bonds, weighted by the offering amount of each bond. (TRACE)
Retail BSI	Daily buy-sell imbalance of retail-sized trades (<\$100k par volume) in target company bonds (TRACE, Mergent, SDC)

Specialist Connection	Indicator variable set to 1 if the insurer transacted with the specialist dealer of a downgraded bond in the 12-month window prior to the downgrade event. We define the specialist dealer as the dealer that intermediates the most trading volume in the bond among all broker-dealers. (S&P)
Specialist Connection Rank	Quintile rank (0 = None, 1 = Low, 5 = High) of the proportion of the specialist-affiliated dealer's transaction volume that is with the insurer in the 12-month window prior to the event, measured in percentage points. We determine the rank of this proportion within the dealer's insurer clients with nonzero trading volume with that dealer within the same window. Specialist Connection Rank is set to zero if Specialist Connection Strength = 0. (S&P)
Specialist Connection Strength to Junk	Proportion of specialist-affiliated dealer's transaction volume that is with the insurer in the 12-month window prior to the event, measured in percentage points. (S&P) Indicator variable set to 1 if the downgrade is a downgrade to junk status, defined as a downgrade from BB- or higher to B+ or lower, by S&P. (Mergent)
Underwriter Connection	Indicator variable set to 1 if the insurer transacted with the issuing underwriter of the downgraded bond in the 12-month window prior to the downgrade event. (S&P, Mergent)
Underwriter Connection Rank	Quintile rank (0 = None, 1 = Low, 5 = High) of the proportion of the underwriter-affiliated dealer's transaction volume that is with the insurer in the 12-month window prior to the event, measured in percentage points. We determine the rank of this proportion within the dealer's insurer clients with nonzero trading volume with that dealer within the same window. Underwriter Connection Rank is set to zero if Underwriter Connection Strength = 0. (S&P, Mergent)
Underwriter Connection Strength	Proportion of underwriter-affiliated dealer's transaction volume that is with the insurer in the 12-month window prior to the event, measured in percentage points. (S&P, Mergent)

Appendix B. Risk-based capital factors

S&P ratings and risk-based capital factors.

Sources: S&P, NAIC.

S&P rating	NAIC designation	RBC factors (%)	
		Life	P&C and health
AAA			
AA+			
AA			
AA-	1	0.39	0.30
A+			
A			
A-			
BBB+			
BBB	2	1.26	1.00
BBB-			
BB+			
BB	3	4.46	2.00
BB-			
B+			
B	4	9.70	4.50
B-			
CCC+			
CCC	5	22.31	10.00
CCC-			
CC	6	30.00	30.00

This table summarizes the risk-based capital (RBC) requirement for corporate bond holdings of insurance companies for the fiscal year 2021. The RBC factors (i.e., the capital charges) depend on the NAIC designation of the security, which is determined by its credit rating. The factors differ between life insurers and property & casualty (P&C) and health insurers.

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